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Regularity estimates in Hölder spaces for Schrödinger operators via a $T1$ theorem

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Abstract We derive Hölder regularity estimates for operators associated with a time-independent Schrödinger operator of the form $-\Delta + V$. The results are obtained by checking a certain condition on the function $T1$. Our general method applies to get regularity estimates for maximal operators and square functions of the heat and Poisson semigroups, for Laplace transform type multipliers and also for Riesz transforms and negative powers $(-\Delta + V)^{-\gamma/2}$, all of them in a unified way.

Keywords Schrödinger operators · Regularity estimates · Campanato spaces · $T1$ criterion · BMO spaces

Mathematics Subject Classification (2000) 35J10 · 35B65 · 26A33 · 42B37 · 46E35 · 42B25

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1 Introduction and statement of the results

Regularity estimates for second-order differential operators are central in the theory of PDEs. In this context, Sobolev and Schauder estimates are fundamental results. The latter can be seen as boundedness between Hölder spaces of negative powers of operators.

In this paper we study regularity estimates in the Hölder classes $C^{0,\alpha}_{\mathcal{L}}$, $0 < \alpha < 1$, of operators associated with the time-independent Schrödinger operator in $\mathbb{R}^n$, $n \geq 3$,

$$\mathcal{L} := -\Delta + V.$$

The nonnegative potential V satisfies a reverse Hölder inequality for some $q \geq n/2$, see Sect. 2.

It is well known that the classical Hölder space $C^{\alpha}(\mathbb{R}^n)$ can be identified with the Campanato space BMO^{α}, see [8]. In the Schrödinger case the analogous result was proved by Bongioanni et al. in [6]. They identified the Hölder space associated with $\mathcal{L}$ with a Campanato type $BMO^{\alpha}_{\mathcal{L}}$ space, see Proposition 2.4 below. Therefore, in order to study regularity estimates, we can take advantage of this characterization. In fact we shall present our results as boundedness of operators between $BMO^{\alpha}_{\mathcal{L}}$ spaces.

The main point of this paper is to give a simple $T1$ criterion for boundedness in $BMO^{\alpha}_{\mathcal{L}}$ of the so-called γ-Schrödinger–Calderón–Zygmund operators T, see Definition 3.1. The advantage of this criterion is that everything is reduced to checking a certain condition on the function $T1$. The method is applied to the maximal operators associated with the semigroups $e^{-t\mathcal{L}}$ and $e^{-t\mathcal{L}^{1/2}}$ (or more general Poisson operators associated with the extension problem for $\mathcal{L}^{\sigma}$), the $\mathcal{L}$-square functions, the Laplace transform type multipliers $m(\mathcal{L})$, the $\mathcal{L}$-Riesz transforms and the negative powers $\mathcal{L}^{-\gamma/2}$, $\gamma > 0$.

We use the notation $f_B = \dfrac{1}{|B|}\displaystyle\int_B f$. The first result reads as follows.

Theorem 1.1 ($T1$ criterion for $BMO^{\alpha}_{\mathcal{L}}$, $0 < \alpha < 1$) *Let T be a γ-Schrödinger–Calderón–Zygmund operator, $\gamma \geq 0$, with smoothness exponent δ, such that $\alpha + \gamma < \min\{1, \delta\}$. Then T is bounded from $BMO^{\alpha}_{\mathcal{L}}$ into $BMO^{\alpha+\gamma}_{\mathcal{L}}$ if and only if there exists a constant C such that*

$$\left(\frac{\rho(x)}{s}\right)^{\alpha} \frac{1}{|B|^{1+\frac{\gamma}{n}}} \int_B |T1(y) - (T1)_B|\, dy \leq C,$$

for every ball $B = B(x, s)$, $x \in \mathbb{R}^n$ and $0 < s \leq \frac{1}{2}\rho(x)$. Here $\rho(x)$ is defined in (2.1).

We can also consider the endpoint case $\alpha = 0$.

Theorem 1.2 ($T1$ criterion for $BMO_{\mathcal{L}}$) *Let T be a γ-Schrödinger–Calderón–Zygmund operator, $0 \leq \gamma < \min\{1, \delta\}$, with smoothness exponent δ. Then T is a bounded operator from $BMO_{\mathcal{L}}$ into $BMO^{\gamma}_{\mathcal{L}}$ if and only if there exists a constant C such that*

$$\log\left(\frac{\rho(x)}{s}\right) \frac{1}{|B|^{1+\frac{\gamma}{n}}} \int_B |T1(y) - (T1)_B|\, dy \leq C,$$

for every ball $B = B(x, s)$, $x \in \mathbb{R}^n$ and $0 < s \leq \frac{1}{2}\rho(x)$.

Observe that for any $x \in \mathbb{R}^n$ and $0 < \alpha \leq 1$, if $0 < s \leq \frac{1}{2}\rho(x)$ then $1 + \log\frac{\rho(x)}{s} \sim \log\frac{\rho(x)}{s}$ and $1 + \dfrac{2^{\alpha}\left(\left(\frac{\rho(x)}{s}\right)^{\alpha} - 1\right)}{2^{\alpha} - 1} \sim \left(\frac{\rho(x)}{s}\right)^{\alpha}$. Therefore, by tracking down the exact constants in the proof, we can see that Theorem 1.2 is indeed the limit case of Theorem 1.1.

Theorem 1.2 is a generalization of the $T1$-type criterion given in Betancor et al. [2] for the case of the harmonic oscillator $H = -\Delta + |x|^2$. Here we require the dimension to be $n \geq 3$, while in Betancor et al. [2] the dimension can be any $n \geq 1$.

As a by-product of our main results we are able to characterize pointwise multipliers of the spaces $BMO_{\mathcal{L}}^{\alpha}$, see Proposition 3.2 below. For pointwise multipliers of the classical BMO^{α} spaces see the papers by Bloom [3], Janson [18] and Nakai and Yabuta [21].

Next we present the announced applications. For the definitions of the operators see Sects. 4.1–4.5.

Theorem 1.3 *Let $0 \leq \alpha < \min\{1, 2 - \frac{n}{q}\}$. The maximal operators associated with the heat semigroup $\{\mathcal{W}_t\}_{t>0}$ and with the generalized Poisson operators $\{\mathcal{P}_t^{\sigma}\}_{t>0}$, the Littlewood–Paley g-functions given in terms of the heat and the Poisson semigroups, and the Laplace transform type multipliers $m(\mathcal{L})$ are bounded from $BMO_{\mathcal{L}}^{\alpha}$ into itself.*

In [11] it was proved that the maximal operator of the heat semigroup, the maximal operator of the Poisson semigroup and the square function of the heat semigroup are bounded in $BMO_{\mathcal{L}}$ and that the fractional integral $\mathcal{L}^{-\gamma/2}$ maps $L^{n/\gamma}(\mathbb{R}^n)$ into $BMO_{\mathcal{L}}$, $0 < \gamma < n$. The square function was also studied in [1]. In [25] it was proved that the fractional integral in the case of the harmonic oscillator has similar boundedness properties in the scale of spaces BMO_H^{α}, or more generally, $C_H^{k,\alpha}(\mathbb{R}^n)$.

The Riesz transforms associated with $\mathcal{L}$ were introduced and studied in $L^p(\mathbb{R}^n)$ in the seminal paper by Shen [22]. Their mapping properties on $BMO_{\mathcal{L}}^{\alpha}$ were developed by Bongioanni et al. in [5]. They also studied the corresponding boundedness results for the negative powers, see [6], and L^p-boundedness for the commutators with a function, see [4]. Following the pattern of the proof of Theorem 1.3, we can recover the results from [5] and [6]. We state them as a theorem for further reference.

Theorem 1.4 *Let $\alpha \geq 0$ and $0 < \gamma < n$. Then:*

- *The $\mathcal{L}$-Riesz transforms are bounded from $BMO_{\mathcal{L}}^{\alpha}$ into itself, for any $0 \leq \alpha < 1 - \frac{n}{q}$, with $q > n$.*
- *The negative powers $\mathcal{L}^{-\gamma/2}$ are bounded from $BMO_{\mathcal{L}}^{\alpha}$ into $BMO_{\mathcal{L}}^{\alpha+\gamma}$ for $\alpha + \gamma < \min\{1, 2 - \frac{n}{q}\}$.*

Regarding Sobolev estimates, more general operators can be considered by replacing $-\Delta$ by some second-order elliptic operator A with bounded measurable coefficients. When A is a degenerate divergence form elliptic operator, some estimates for the Green function and the heat semigroup were obtained by Dziubański in [10]. A priori L^p estimates and global existence and uniqueness results in L^p for the case when A is in nondivergence form with VMO coefficients were found by Bramanti et al. in [7].

The use of the action of an operator T on the function 1 in order to get some boundedness properties of T goes back to the celebrated work by G. David and J.-L. Journé, see [9]. For vector-valued versions of these criteria, see the papers by Hytönen [16] and Hytönen and Weis [17].

The paper is organized as follows. In Sect. 2 we collect the technical results about the space $BMO_{\mathcal{L}}^{\alpha}$. Section 3 is devoted to the proofs of the main theorems. The applications are given in Sect. 4. Through the paper the letters C and c denote positive constants that may change at each occurrence, and $\mathcal{S}$ is the class of rapidly decreasing C^{∞} functions in $\mathbb{R}^n$.

2 The spaces $BMO^{\alpha}_{\mathcal{L}}$, $0 \le \alpha \le 1$

The nonnegative potential V satisfies a reverse Hölder inequality for some $q \ge \frac{n}{2}$; that is, there exists a constant $C = C(q, V)$ such that

$$\left(\frac{1}{|B|}\int_B V(y)^q \, dy\right)^{1/q} \le \frac{C}{|B|}\int_B V(y)\, dy,$$

for all balls $B \subset \mathbb{R}^n$. We write $V \in RH_q$. Associated with this potential, Shen defines the critical radii function in [22] as

$$\rho(x) := \sup\left\{r > 0 : \frac{1}{r^{n-2}}\int_{B(x,r)} V(y)\, dy \le 1\right\}, \quad x \in \mathbb{R}^n. \tag{2.1}$$

We have $0 < \rho(x) < \infty$.

Let us begin with some properties of the critical radii function ρ.

Lemma 2.1 (See [22, Lemma 1.4]) *There exist $c > 0$ and $k_0 \ge 1$ such that for all $x, y \in \mathbb{R}^n$,*

$$c^{-1}\rho(x)\left(1 + \frac{|x-y|}{\rho(x)}\right)^{-k_0} \le \rho(y) \le c\rho(x)\left(1 + \frac{|x-y|}{\rho(x)}\right)^{\frac{k_0}{k_0+1}}.$$

In particular, there exists a positive constant $C_1 < 1$ such that if $|x-y| \le \rho(x)$ then $C_1\rho(x) < \rho(y) < C_1^{-1}\rho(x)$.

Covering by critical balls. According to [12, Lemma 2.3] there exists a sequence of points $\{x_k\}_{k=1}^{\infty}$ in $\mathbb{R}^n$ such that if $Q_k := B(x_k, \rho(x_k))$, $k \in \mathbb{N}$, then

(a) $\cup_{k=1}^{\infty} Q_k = \mathbb{R}^n$, and
(b) there exists $N \in \mathbb{N}$ such that card $\{j \in \mathbb{N} : Q_j^{**} \cap Q_k^{**} \ne \emptyset\} \le N$, for every $k \in \mathbb{N}$.

For a ball B, the notation B^* above means the ball with the same center as B and twice the radius.

The definition of space $BMO_{\mathcal{L}}$ was given in Dziubański et al. [11]. The space $BMO^{\alpha}_{\mathcal{L}}$, $0 < \alpha \le 1$, was introduced in Bongioanni et al. [6]. We collect from there the following facts.

A locally integrable function f in $\mathbb{R}^n$ is in $BMO^{\alpha}_{\mathcal{L}}$, $0 \le \alpha \le 1$ provided there exists $C > 0$ such that

(i) $\frac{1}{|B|}\int_B |f(x) - f_B|\, dx \le C\,|B|^{\frac{\alpha}{n}}$, for every ball B in $\mathbb{R}^n$, and
(ii) $\frac{1}{|B|}\int_B |f(x)|\, dx \le C\,|B|^{\frac{\alpha}{n}}$, for every $B = B(x_0, r_0)$, where $x_0 \in \mathbb{R}^n$ and $r_0 \ge \rho(x_0)$.

The norm $\|f\|_{BMO^{\alpha}_{\mathcal{L}}}$ of f is defined as the minimum $C > 0$ such that (i) and (ii) above hold. We have $BMO^0_{\mathcal{L}} = BMO_{\mathcal{L}}$.

By using the classical John–Nirenberg inequality, it can be seen that if in (i) and (ii) L^1-norms are replaced by L^p-norms, for $1 < p < \infty$, then the space $BMO^{\alpha}_{\mathcal{L}}$ does not change and equivalent norms appear. In this case the conditions read:

(i)$_p$ $\left(\frac{1}{|B|}\int_B |f(x)-f_B|^p\ \mathrm{d}x\right)^{1/p} \le C\,|B|^{\frac{\alpha}{n}}$, for every ball B in $\mathbb{R}^n$, and

(ii)$_p$ $\left(\frac{1}{|B|}\int_B |f(x)|^p\ \mathrm{d}x\right)^{1/p} \le C\,|B|^{\frac{\alpha}{n}}$, for every $B = B(x_0, r_0)$, where $x_0 \in \mathbb{R}^n$ and $r_0 \ge \rho(x_0)$.

Let us note that if (ii) (resp. (ii)$_p$) above is true for some ball B then (i) (resp. (i)$_p$) holds for the same ball, so we might ask for (i) (resp. (i)$_p$) only for balls with radii smaller than $\rho(x)$.

The restriction $\alpha \le 1$ in the definition above is necessary because if $\alpha > 1$ then the space $BMO^\alpha_{\mathcal{L}}$ only contains constant functions.

Proposition 2.2 *Let $B = B(x, r)$ with $r < \rho(x)$.*

(1) *(See* [11, Lemma 2]*) If $f \in BMO_{\mathcal{L}}$ then $|f_B| \le C\left(1 + \log\frac{\rho(x)}{r}\right)\|f\|_{BMO_{\mathcal{L}}}$.*
(2) *(See* [20, *Proposition* 4.3]*) If $f \in BMO^\alpha_{\mathcal{L}}$, $0 < \alpha \le 1$, then we have $|f_B| \le C_\alpha \|f\|_{BMO^\alpha_{\mathcal{L}}}\rho(x)^\alpha$.*
(3) *(See* [6, *Proposition* 3] *) A function f belongs to $BMO^\alpha_{\mathcal{L}}$, $0 \le \alpha \le 1$, if and only if f satisfies* (i) *for every ball $B = B(x_0, r_0)$ with $r_0 < \rho(x_0)$ and $|f|_{Q_k} \le C\,|Q_k|^{1+\frac{\alpha}{n}}$, for all balls Q_k given in the covering by critical balls above.*

Lemma 2.3 (Boundedness criterion) *Let S be a linear operator defined on $BMO^\alpha_{\mathcal{L}}$, $0 \le \alpha \le 1$. Then S is bounded from $BMO^\alpha_{\mathcal{L}}$ into $BMO^{\alpha+\gamma}_{\mathcal{L}}$, $\alpha + \gamma \le 1$, $\gamma \ge 0$, if there exists $C > 0$ such that, for every $f \in BMO^\alpha_{\mathcal{L}}$ and $k \in \mathbb{N}$,*

(A_k) $\dfrac{1}{|Q_k|^{1+\frac{\alpha+\gamma}{n}}}\displaystyle\int_{Q_k} |Sf(x)|\,dx \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}$, *and*
(B_k) $\|Sf\|_{BMO^{\alpha+\gamma}(Q_k^*)} \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}$, *where $BMO^\alpha(Q_k^*)$ denotes the usual BMO^α space on the ball Q_k^*.*

Proof For $\alpha = 0$ the result is already contained in Dziubański et al. [11, p. 346]. The general statement follows immediately from the definition of $BMO^\alpha_{\mathcal{L}}$ and Lemma 2.1 (see Proposition 2.2). □

The duality of the $\mathcal{L}$-Hardy space $H^1_{\mathcal{L}}$ with $BMO_{\mathcal{L}}$ was proved in Dziubański et al. [11]. As already mentioned in the paper by Bongioanni et al. [6], the $BMO^\alpha_{\mathcal{L}}$ spaces are the duals of the $H^p_{\mathcal{L}}$ spaces defined in Dziubański and Zienkiewicz [12–14]. In fact, if $s > n$ and $0 \le \alpha < 1$ then the dual of $H^{\frac{n}{n+\alpha}}_{\mathcal{L}}$ is $BMO^\alpha_{\mathcal{L}}$; see also [15], references in [20] and [26].

We denote by $C^\alpha(\mathbb{R}^n)$ the space of α-Hölder continuous functions on $\mathbb{R}^n$ and by $[f]_{C^\alpha}$ its usual seminorm. Recall that $BMO^\alpha(\mathbb{R}^n) = C^\alpha(\mathbb{R}^n)$ with $\|f\|_{BMO^\alpha(\mathbb{R}^n)} \sim [f]_{C^\alpha}$.

Proposition 2.4 (Campanato description, [6, Proposition 4]) *Let $0 < \alpha \le 1$. A function f belongs to $BMO^\alpha_{\mathcal{L}}$ if and only if $f \in C^\alpha(\mathbb{R}^n)$ and $|f(x)| \le C\rho(x)^\alpha$, for all $x \in \mathbb{R}^n$. Moreover, $\|f\|_{BMO^\alpha_{\mathcal{L}}} \sim [f]_{C^\alpha(\mathbb{R}^n)} + \|f\rho^{-\alpha}\|_{L^\infty(\mathbb{R}^n)}$.*

In the following lemma we present examples of families of functions indexed by $x_0 \in \mathbb{R}^n$ and $0 < s \le \rho(x_0)$ that are uniformly bounded in $BMO^\alpha_{\mathcal{L}}$. They will be useful in the sequel.

Lemma 2.5 *There exist constants $C, C_\alpha > 0$ such that for every $x_0 \in \mathbb{R}^n$ and $0 < s \le \rho(x_0)$,*

(a) *the function*

$$g_{x_0,s}(x) := \chi_{[0,s]}(|x-x_0|)\log\left(\frac{\rho(x_0)}{s}\right) + \chi_{(s,\rho(x_0))]}(|x-x_0|)\log\left(\frac{\rho(x_0)}{|x-x_0|}\right),$$

$x \in \mathbb{R}^n$, *belongs to $BMO_{\mathcal{L}}$ and $\|g_{x_0,s}\|_{BMO_{\mathcal{L}}} \le C$;*

(b) *the function*

$$f_{x_0,s}(x) = \chi_{[0,s]}(|x-x_0|)\left(\rho(x_0)^\alpha - s^\alpha\right) + \chi_{(s,\rho(x_0))]}(|x-x_0|)\left(\rho(x_0)^\alpha - |x-x_0|^\alpha\right),$$

$x \in \mathbb{R}^n$, *belongs to $BMO^\alpha_{\mathcal{L}}$, $0 < \alpha \le 1$, and $\|f_{x_0,s}\|_{BMO^\alpha_{\mathcal{L}}} \le C_\alpha$.*

Proof The proof of (a) follows the same lines as the proof of Lemma 2.1 in Betancor et al. [2]. We omit the details.

Let us continue with (b). Recall that the function $h(x) = (1-|x|^\alpha)\,\chi_{[0,1]}(|x|)$ is in $BMO^\alpha(\mathbb{R}^n)$. Hence, for every $R > 0$, the function $h_R(x) := R^\alpha h(x/R)$ is in $BMO^\alpha(\mathbb{R}^n)$ and $\|h_R\|_{BMO^\alpha(\mathbb{R}^n)} \le C$, where $C > 0$ is independent of R. Moreover, for every $R > 0$ and $S \ge 1$, the function $h_{R,S}(x) = \min\{R^\alpha(1-S^{-\alpha}), R^\alpha h(x/R)\}$ belongs to $BMO^\alpha(\mathbb{R}^n)$ and $\|h_{R,S}\|_{BMO^\alpha(\mathbb{R}^n)} \le C$, where $C > 0$ does not depend on R and S. Then, since for every $x_0 \in \mathbb{R}^n$ and $0 < s \le \rho(x_0)$,

$$f_{x_0,s}(x) = h_{\rho(x_0),\frac{\rho(x_0)}{s}}(x-x_0), \quad x \in \mathbb{R}^n,$$

we get $f_{x_0,s} \in BMO^\alpha(\mathbb{R}^n) = C^\alpha(\mathbb{R}^n)$ and $\|f_{x_0,s}\|_{BMO^\alpha(\mathbb{R}^n)} \le C$. This, the obvious inequality $\left|f_{x_0,s}(x)\right| \le C\rho(x)^\alpha$, for all x, uniformly in x_0 and $s \le \rho(x_0)$, and Proposition 2.4 imply the conclusion. □

3 Operators and proofs of the main results

3.1 The operators related to $\mathcal{L}$

We denote by $L^p_c(\mathbb{R}^n)$ the set of functions $f \in L^p(\mathbb{R}^n)$, $1 \le p \le \infty$, whose support $\mathrm{supp}(f)$ is a compact subset of $\mathbb{R}^n$.

Definition 3.1 Let $0 \le \gamma < n$, $1 < p \le q < \infty$, $\frac{1}{q} = \frac{1}{p} - \frac{\gamma}{n}$. Let T be a bounded linear operator from $L^p(\mathbb{R}^n)$ into $L^q(\mathbb{R}^n)$ such that

$$Tf(x) = \int_{\mathbb{R}^n} K(x,y)f(y)\,\mathrm{d}y, \quad f \in L^p_c(\mathbb{R}^n) \text{ and a.e. } x \notin \mathrm{supp}(f).$$

We shall say that T is a γ-Schrödinger–Calderón–Zygmund operator with regularity exponent $\delta > 0$ if for some constant C

(1) $|K(x,y)| \le \dfrac{C}{|x-y|^{n-\gamma}}\left(1 + \dfrac{|x-y|}{\rho(x)}\right)^{-N}$, for all $N > 0$ and $x \ne y$,

(2) $|K(x,y)-K(x,z)| + |K(y,x)-K(z,x)| \le C\dfrac{|y-z|^\delta}{|x-y|^{n-\gamma+\delta}}$, when $|x-y| > 2|y-z|$.

Definition of Tf for $f \in BMO^\alpha_{\mathcal{L}}, 0 \le \alpha \le 1$. Suppose that $f \in BMO^\alpha_{\mathcal{L}}$ and $R \ge \rho(x_0)$, $x_0 \in \mathbb{R}^n$. We define

$$Tf(x) = T\left(f\chi_{B(x_0,R)}\right)(x) + \int\limits_{B(x_0,R)^c} K(x,y)f(y)\,\mathrm{d}y, \quad \text{a.e. } x \in B(x_0,R).$$

Note that the first term in the right-hand side makes sense since $f\chi_{B(x_0,R)} \in L^p_c(\mathbb{R}^n)$. The integral in the second term is absolutely convergent. Indeed, by Lemma 2.1, there exists a constant C such that for any $x \in B(x_0,R)$,

$$\begin{aligned}\rho(x) &\le c\rho(x_0)\left(1+\frac{|x-x_0|}{\rho(x_0)}\right)^{\frac{k_0}{k_0+1}} \le C\left(\rho(x_0)+\rho(x_0)^{1-\frac{k_0}{k_0+1}}|x-x_0|^{\frac{k_0}{k_0+1}}\right)\\ &\le C\left(R+R^{1-\frac{k_0}{k_0+1}}|x-x_0|^{\frac{k_0}{k_0+1}}\right) \le C2R.\end{aligned}$$

Hence, using the γ-Schrödinger–Calderón–Zygmund condition (1) for K with $N-\gamma > \alpha$,

$$\begin{aligned}\int\limits_{B(x_0,2R)^c} |K(x,y)||f(y)|\,\mathrm{d}y &\le C\sum_{j=1}^{\infty}\int\limits_{2^jR<|y-x_0|\le 2^{j+1}R} \frac{\rho(x)^N}{|x-y|^{n+N-\gamma}}|f(y)|\,\mathrm{d}y\\ &\le C\sum_{j=1}^{\infty}\frac{\rho(x)^N}{(2^jR-R)^{n+N-\gamma}}\int\limits_{|y-x_0|\le 2^{j+1}R}|f(y)|\,\mathrm{d}y\\ &\le CR^{\alpha+\gamma}\|f\|_{BMO^\alpha_{\mathcal{L}}}, \quad \text{a.e. } x \in B(x_0,R). \qquad (3.1)\end{aligned}$$

The definition of $Tf(x)$ is also independent of R in the sense that if $B(x_0,R) \subset B(x_0',R')$, with $R' \ge \rho(x_0)$, then the definition using $B(x_0',R')$ coincides almost everywhere in $B(x_0,R)$ with the one just given, because in that situation,

$$\begin{aligned}&T\left(f\chi_{B(x_0',R')}\right)(x) - T\left(f\chi_{B(x_0,R)}\right)(x)\\ &= T\left(f\chi_{B(x_0',R')\setminus B(x_0,R)}\right)(x) = \int\limits_{B(x_0',R')\setminus B(x_0,R)} K(x,y)f(y)\,\mathrm{d}y\\ &= \int\limits_{B(x_0,R)^c} K(x,y)f(y)\,\mathrm{d}y - \int\limits_{B(x_0',R')^c} K(x,y)f(y)\,\mathrm{d}y,\end{aligned}$$

for almost every $x \in B(x_0,R)$.

The definition just given above is equally valid for $f \equiv 1 \in BMO_{\mathcal{L}}$.

Next we derive an expression for Tf, where $T1$ appears, that will be useful in the proof of our main results. Let $x_0 \in \mathbb{R}^n$ and $r_0 > 0$. For $B = B(x_0,r_0)$ we clearly have

$$f = (f-f_B)\chi_{B^{***}} + (f-f_B)\chi_{(B^{***})^c} + f_B =: f_1 + f_2 + f_3. \qquad (3.2)$$

Let us choose $R \ge \rho(x_0)$ such that $B^{***} \subset B(x_0,R)$. By using the definition of Tf given above, the identity in (3.2), adding and subtracting f_B in the integral over $B(x_0,R)^c$ and collecting terms we get

$$
\begin{aligned}
Tf(x) &= T\left(f\chi_{B(x_0,R)}\right)(x) + \int_{B(x_0,R)^c} K(x,y)f(y)\,\mathrm{d}y \\
&= T\left((f-f_B)\chi_{B^{***}}\right)(x) + T\left((f-f_B)\chi_{B(x_0,R)\setminus B^{***}}\right)(x) \\
&\quad + f_B T\left(\chi_{B(x_0,R)}\right)(x) \\
&\quad + \int_{B(x_0,R)^c} K(x,y)(f(y)-f_B)\,\mathrm{d}y + f_B \int_{B(x_0,R)^c} K(x,y)\,\mathrm{d}y \\
&= T\left((f-f_B)\chi_{B^{***}}\right)(x) + \int_{(B^{***})^c} K(x,y)(f(y)-f_B)\,\mathrm{d}y \\
&\quad + f_B T1(x), \quad \text{a.e. } x \in B^{***}.
\end{aligned} \tag{3.3}
$$

We observe that there exists a constant C such that

$$
\frac{1}{|B|^{1+\frac{\gamma}{n}}} \int_B |T1(y)|\,\mathrm{d}y \le C, \quad \text{for all } B = B(x,\rho(x)),\ x \in \mathbb{R}^n. \tag{3.4}
$$

Indeed, by Hölder's inequality and the $L^p - L^q$ boundedness of T,

$$
\begin{aligned}
\frac{1}{|B|^{1+\frac{\gamma}{n}}} \int_B |T\left(\chi_{B^*}\right)(y)|\,\mathrm{d}y &\le \frac{1}{|B|^{\frac{1}{q}+\frac{\gamma}{n}}} \left(\int_B |T\left(\chi_{B^*}\right)(y)|^q\,\mathrm{d}y\right)^{1/q} \\
&\le C\frac{|B|^{1/p}}{|B|^{\frac{1}{q}+\frac{\gamma}{n}}} = C.
\end{aligned}
$$

By the integral representation of T and the size condition (1) on K with $N = n+\gamma$, for $y \in B(x,\rho(x))$ we have

$$
\begin{aligned}
\left|T\left(\chi_{(B^*)^c}\right)(y)\right| &\le C\sum_{k=1}^{\infty} \int_{2^j\rho(x)\le|x-z|<2^{j+1}\rho(x)} \frac{\rho(y)^{n+\gamma}}{|y-z|^{2n}}\,\mathrm{d}z \\
&\le C\rho(y)^{n+\gamma}\sum_{k=1}^{\infty} \frac{(2^{j+1}\rho(x))^n}{(2^j\rho(x)-\rho(x))^{2n}} \le C\rho(x)^{\gamma},
\end{aligned}
$$

because $\rho(x) \sim \rho(y)$. Thus (3.4) follows by linearity.

3.2 Proofs of Theorems 1.1 and 1.2

Proof of Theorem 1.1 First we shall see that the condition on $T1$ implies that T is bounded from $BMO^{\alpha}_{\mathcal{L}}$ into $BMO^{\alpha+\gamma}_{\mathcal{L}}$. In order to do this, we will show that there exists $C > 0$ such that the properties (A_k) and (B_k) stated in Lemma 2.3 hold for every $k \in \mathbb{N}$ and $f \in BMO^{\alpha}_{\mathcal{L}}$.

We begin with (A_k). According to (3.3) with $B = Q_k$,

$$
\begin{aligned}
Tf(x) &= T\left((f-f_{Q_k})\chi_{Q_k^{***}}\right)(x) + \int_{(Q_k^{***})^c} K(x,y)(f(y)-f_{Q_k})\,\mathrm{d}y \\
&\quad + f_{Q_k}T1(x), \quad \text{a.e. } x \in Q_k.
\end{aligned}
$$

As T maps $L^p(\mathbb{R}^n)$ into $L^q(\mathbb{R}^n)$, $\frac{1}{q} = \frac{1}{p} - \frac{\gamma}{n}$, by Hölder's inequality,

$$\begin{aligned}
&\frac{1}{|Q_k|^{1+\frac{\alpha+\gamma}{n}}} \int_{Q_k} \left| T\left((f - f_{Q_k})\chi_{Q_k^{***}}\right)(x)\right| \mathrm{d}x \\
&\quad \le \frac{1}{|Q_k|^{\frac{1}{q}+\frac{\alpha+\gamma}{n}}} \left(\int_{Q_k} \left| T\left((f - f_{Q_k})\chi_{Q_k^{***}}\right)(x)\right|^q \mathrm{d}x \right)^{1/q} \\
&\quad \le \frac{C}{|Q_k|^{\frac{\alpha}{n}}} \left(\frac{1}{|Q_k|} \int_{Q_k^{***}} \left| f(x) - f_{Q_k}\right|^p \mathrm{d}x \right)^{1/p} \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}.
\end{aligned}$$

On the other hand, given $x \in Q_k$, we have $\rho(x) \sim \rho(x_k)$ and if $|x_k - y| > 2^j \rho(x_k)$, $j \in \mathbb{N}$, then $|x - y| \ge 2^{j-1}\rho(x_k)$. By the size condition (1) of the kernel K, for any $N > \alpha$ we have

$$\begin{aligned}
&\frac{1}{|Q_k|^{\frac{\alpha+\gamma}{n}}} \left| \int_{(Q_k^{***})^c} K(x,y)\big(f(y) - f_{Q_k}\big)\,\mathrm{d}y \right| \\
&\quad \le \frac{1}{|Q_k|^{\frac{\alpha+\gamma}{n}}} \int_{(Q_k^{***})^c} |K(x,y)|\left|f(y) - f_{Q_k}\right|\,\mathrm{d}y \\
&\quad \le \frac{C}{|Q_k|^{\frac{\alpha+\gamma}{n}}} \int_{(Q_k^{***})^c} \frac{1}{|x-y|^{n-\gamma}} \left(1 + \frac{|x-y|}{\rho(x)}\right)^{-N} \left|f(y) - f_{Q_k}\right|\,\mathrm{d}y \\
&\quad \le \frac{C}{|Q_k|^{\frac{\alpha+\gamma}{n}}} \sum_{j=3}^{\infty} \int_{2^j\rho(x_k) < |x_k - y| \le 2^{j+1}\rho(x_k)} \frac{\rho(x)^N}{|x-y|^{n-\gamma+N}} \left|f(y) - f_{Q_k}\right|\,\mathrm{d}y \\
&\quad \le \frac{C}{\rho(x_k)^\alpha} \sum_{j=3}^{\infty} \frac{\rho(x_k)^N}{\left(2^j\rho(x_k)\right)^{n+N}} \int_{|x_k - y| \le 2^{j+1}\rho(x_k)} \left|f(y) - f_{Q_k}\right|\,\mathrm{d}y \\
&\quad \le C \sum_{j=3}^{\infty} 2^{-j(N-\alpha)}(j+1)\|f\|_{BMO^\alpha_{\mathcal{L}}} \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}.
\end{aligned}$$

Finally, by (3.4),

$$\frac{1}{|Q_k|^{1+\frac{\alpha+\gamma}{n}}} \int_{Q_k} \left|f_{Q_k} T1(x)\right|\,\mathrm{d}x = \frac{|f_{Q_k}|}{|Q_k|^{\frac{\alpha}{n}}} \frac{1}{|Q_k|^{1+\frac{\gamma}{n}}} \int_{Q_k} |T1(x)|\,\mathrm{d}x \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}.$$

Hence, we conclude that (A_k) holds for T with a constant C that does not depend on k.

Let us continue with (B_k). Let $B = B(x_0, r_0) \subseteq Q_k^*$, where $x_0 \in \mathbb{R}^n$ and $r_0 > 0$. Note that if $r_0 \ge \frac{1}{2}\rho(x_0)$ then $\rho(x_0) \sim \rho(x_k) \sim r_0$, so proceeding as above we have

$$\frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}} \int_B |Tf(x) - (Tf)_B|\,\mathrm{d}x \le \frac{2}{|B|^{1+\frac{\alpha+\gamma}{n}}} \int_B |Tf(x)|\,\mathrm{d}x \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}.$$

Assume next that $0 < r_0 < \frac{1}{2}\rho(x_0)$. We have

$$\begin{aligned}
&\frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B |Tf(x)-(Tf)_B|\,\mathrm{d}x\\
&\le \frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B \frac{1}{|B|}\int_B |Tf_1(x)-Tf_1(z)|\,\mathrm{d}z\,\mathrm{d}x\\
&\quad+\frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B \frac{1}{|B|}\int_B |F_2(x)-F_2(z)|\,\mathrm{d}z\,\mathrm{d}x\\
&\quad+\frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B |Tf_3(x)-(Tf_3)_B|\,\mathrm{d}x =: L_1+L_2+L_3,
\end{aligned}$$

where $f = f_1 + f_2 + f_3$ as in (3.2) and we defined

$$F_2(x) = \int_{(B^{***})^c} K(x,y) f_2(y)\,\mathrm{d}y, \quad x \in B.$$

Again Hölder's inequality and $L^p - L^q$ boundedness of T give $L_1 \le C\|f\|_{BMO^\alpha_{\mathcal{L}}}$. Let us estimate L_2. Take $x, z \in B$ and $y \in (B^{***})^c$. Then $8r_0 < |y-x_0| \le |y-x| + r_0$ and therefore $2|x-x_0| < 4r_0 < |y-x|$. Under these conditions we can apply the smoothness of the kernel (recall Definition 3.1(2)) and the restriction $\alpha+\gamma < \min\{1,\delta\}$ to get

$$\begin{aligned}
&\frac{1}{|B|^{\frac{\alpha+\gamma}{n}}}|F_2(x)-F_2(z)| \le \frac{C}{r_0^{\alpha+\gamma}}\int_{(B^{***})^c}|K(x,y)-K(z,y)|\,|f(y)-f_B|\,\mathrm{d}y\\
&\le \frac{C}{r_0^{\alpha+\gamma}}\sum_{j=3}^{\infty}\int_{2^j r_0\le|x_0-y|<2^{j+1}r_0}\frac{|x-z|^\delta}{|x-y|^{n-\gamma+\delta}}|f(y)-f_B|\,\mathrm{d}y\\
&\le \frac{C}{r_0^{\alpha+\gamma}}\sum_{j=3}^{\infty}\frac{r_0^\delta}{((2^j-1)r_0)^{n-\gamma+\delta}}\int_{2^j r_0\le|x_0-y|<2^{j+1}r_0}|f(y)-f_B|\,\mathrm{d}y\\
&\le C\sum_{j=3}^{\infty}\frac{2^{-j(\delta-(\alpha+\gamma))}}{(2^{j+1}r_0)^{n+\alpha}}\int_{|x_0-y|<2^{j+1}r_0}|f(y)-f_B|\,\mathrm{d}y\\
&= C\sum_{j=3}^{\infty}\frac{2^{-j(\delta-(\alpha+\gamma))}}{(2^{j+1}r_0)^{n+\alpha}}\int_{|x_0-y|<2^{j+1}r_0}\Big|f(y)-f_{2^{j+1}B}+\sum_{k=0}^{j}(f_{2^{k+1}B}-f_{2^kB})\Big|\,\mathrm{d}y\\
&\le C\sum_{j=3}^{\infty}2^{-j(\delta-(\alpha+\gamma))}\Bigg[\frac{1}{(2^{j+1}r_0)^{n+\alpha}}\int_{|x_0-y|<2^{j+1}r_0}|f(y)-f_{2^{j+1}B}|\,\mathrm{d}y\\
&\quad+\frac{1}{(2^{j+1}r_0)^{\alpha}}\sum_{k=0}^{j}\frac{|2^{k+1}B|}{|2^kB|}\frac{1}{|2^{k+1}B|}\int_{2^{k+1}B}|f(y)-f_{2^{k+1}B}|\,\mathrm{d}y\Bigg]
\end{aligned}$$

$$\leq C\sum_{j=3}^{\infty} 2^{-j(\delta-(\alpha+\gamma))}\left[\|f\|_{BMO^{\alpha}_{\mathcal{L}}}+\sum_{k=0}^{j}\frac{1}{|2^{k+1}B|^{1+\frac{\alpha}{n}}}\int\limits_{2^{k+1}B}|f(y)-f_{2^{k+1}B}|\,\mathrm{d}y\right]$$

$$\leq C\|f\|_{BMO^{\alpha}_{\mathcal{L}}}\sum_{j=3}^{\infty}2^{-j(\delta-(\alpha+\gamma))}(j+2)=C\|f\|_{BMO^{\alpha}_{\mathcal{L}}}.$$

Therefore, $L_2 \leq C\|f\|_{BMO^{\alpha}_{\mathcal{L}}}$. We finally consider L_3. Using Proposition 2.2(2) and the assumption on $T1$, it follows that

$$\begin{aligned} L_3 &= \frac{|f_B|}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B |T1(x)-(T1)_B|\ \mathrm{d}x \\ &\leq C\|f\|_{BMO^{\alpha}_{\mathcal{L}}}\left(\frac{\rho(x_0)}{r_0}\right)^{\alpha}\frac{1}{|B|^{1+\frac{\gamma}{n}}}\int_B |T1(x)-(T1)_B|\ \mathrm{d}x \\ &\leq C\|f\|_{BMO^{\alpha}_{\mathcal{L}}}. \end{aligned} \tag{3.5}$$

This concludes the proof of (B_k). Hence, T is bounded from $BMO^{\alpha}_{\mathcal{L}}$ into $BMO^{\alpha+\gamma}_{\mathcal{L}}$.

Let us now prove the converse statement. Suppose that T is bounded from $BMO^{\alpha}_{\mathcal{L}}$ into $BMO^{\alpha+\gamma}_{\mathcal{L}}$. Let $x_0 \in \mathbb{R}^n$ and $0 < s \leq \frac{1}{2}\rho(x_0)$ and $B = B(x_0, s)$. For such x_0 and s consider the nonnegative function $f_0(x) \equiv f_{x_0,s}(x)$ defined in Lemma 2.5. Using the decomposition $f_0 = (f_0-(f_0)_B)\chi_{B^{***}} + (f_0-(f_0)_B)\chi_{(B^{***})^c} + (f_0)_B =: f_1 + f_2 + (f_0)_B$ as in (3.2), we can write $(f_0)_B T1(y) = Tf_0(y) - Tf_1(y) - Tf_2(y)$, so

$$(f_0)_B\frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B |T1(y)-(T1)_B|\ \mathrm{d}y \leq \sum_{i=0}^{2}\frac{1}{|B|^{1+\frac{\alpha+\gamma}{n}}}\int_B |Tf_i(y)-(Tf_i)_B|\ \mathrm{d}y.$$

We can check that each of the three terms above is controlled by $C\|f_0\|_{BMO^{\alpha}_{\mathcal{L}}} \leq C$, where C is independent of x_0 and s. Indeed, the case $i = 0$ follows by the hypothesis about the boundedness of T. For $i = 1$ the estimate follows, as usual, by Hölder's inequality and $L^p - L^q$ boundedness of T. The term for $i = 2$ is done as L_2 above. Thus, since $(f_0)_B = C(\rho(x_0)^{\alpha} - s^{\alpha})$ we obtain

$$\left(\frac{\rho(x_0)}{s}\right)^{\alpha}\frac{1}{|B|^{1+\frac{\gamma}{n}}}\int_B |T1(y)-(T1)_B|\ \mathrm{d}y \leq C.$$

□

Proof of Theorem 1.2 The proof is the same as the proof of Theorem 1.1 putting $\alpha = 0$ everywhere, except for just two differences. The first one is the estimate of the term L_3, where we must apply Proposition 2.2(1) instead of (2). The second difference is the proof of the converse, where instead of $f_{x_0,s}(x)$ we have to consider the function $g_{x_0,s}(x)$ of Lemma 2.5. □

3.3 Pointwise multipliers in $BMO^{\alpha}_{\mathcal{L}}$, $0 \leq \alpha < 1$

Proposition 3.2 *Let ψ be a measurable function on $\mathbb{R}^n$. We denote by T_ψ the multiplier operator defined by $T_\psi(f) = f\psi$. Then*

(A) *T_ψ is a bounded operator in $BMO_{\mathcal{L}}$ if and only if $\psi \in L^\infty(\mathbb{R}^n)$ and there exists $C > 0$ such that, for all balls $B = B(x_0, s)$ with $0 < s < \frac{1}{2}\rho(x_0)$,*

$$\log\left(\frac{\rho(x_0)}{s}\right)\frac{1}{|B|}\int_B |\psi(y) - \psi_B|\,dy \le C.$$

(B) *T_ψ is a bounded operator in $BMO^\alpha_{\mathcal{L}}$, $0 < \alpha < 1$, if and only if $\psi \in L^\infty(\mathbb{R}^n)$ and there exists $C > 0$ such that, for all balls $B = B(x_0, s)$ with $0 < s < \frac{1}{2}\rho(x_0)$,*

$$\left(\frac{\rho(x_0)}{s}\right)^\alpha \frac{1}{|B|}\int_B |\psi(y) - \psi_B|\,dy \le C.$$

Remark 3.3 If $\psi \in C^{0,\beta}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, $0 < \beta \le 1$, then T_ψ is bounded on $BMO_{\mathcal{L}}$. Moreover, if for some γ-Schrödinger–Calderón–Zygmund operator T we have that $T1$ defines a pointwise multiplier in $BMO^\alpha_{\mathcal{L}}$, then the proposition above and Theorems 1.2 and 1.1 imply that T is a bounded operator on $BMO^\alpha_{\mathcal{L}}$.

Proof of Proposition 3.2 Let us first prove (B). Suppose that T_ψ is a bounded operator on $BMO^\alpha_{\mathcal{L}}$, $0 < \alpha < 1$. For the function $f_{x_0,s}(x)$ defined in Lemma 2.5 and any ball $B = B(x_0, s)$ with $0 < s \le \frac{1}{2}\rho(x_0)$, by Proposition 2.2(2) applied to $f\psi$ and the hypothesis, we get

$$\begin{aligned}
&\left(\frac{\rho(x_0)}{s}\right)^\alpha \frac{1}{|B|}\int_B |\psi(x)|\,\mathrm{d}x \\
&\le C_\alpha \frac{(\rho(x_0)^\alpha - s^\alpha)}{|B|^{1+\frac{\alpha}{n}}}\int_B |\psi(x)|\,\mathrm{d}x = \frac{C_\alpha}{|B|^{1+\frac{\alpha}{n}}}\int_B |\psi(x) f_{x_0,s}(x)|\,\mathrm{d}x \\
&\le \frac{C_\alpha}{|B|^{1+\frac{\alpha}{n}}}\int_B |(\psi f_{x_0,s})(x) - (\psi f_{x_0,s})_B|\,\mathrm{d}x + \frac{C_\alpha}{|B|^{\frac{\alpha}{n}}}(\psi f_{x_0,s})_B \\
&\le C_\alpha \|f_{x_0,s}\|_{BMO^\alpha_{\mathcal{L}}} + C_\alpha\left(\frac{\rho(x_0)}{s}\right)^\alpha \|\psi f_{x_0,s}\|_{BMO^\alpha_{\mathcal{L}}} \\
&\le C_\alpha\left(\frac{\rho(x_0)}{s}\right)^\alpha \|f_{x_0,s}\|_{BMO^\alpha_{\mathcal{L}}} \le C\left(\frac{\rho(x_0)}{s}\right)^\alpha.
\end{aligned}$$

Hence, $|\psi|_B \le C$ with C independent of B, so that ψ is bounded. Next we check the condition on ψ. We have

$$\begin{aligned}
&\left(\frac{\rho(x_0)}{s}\right)^\alpha \frac{1}{|B|}\int_B |\psi(x) - \psi_B|\,\mathrm{d}x \le C_\alpha \frac{(\rho(x_0)^\alpha - s^\alpha)}{|B|^{1+\frac{\alpha}{n}}}\int_B |\psi(x) - \psi_B|\,\mathrm{d}x \\
&\le \frac{C_\alpha}{|B|^{1+\frac{\alpha}{n}}}\int_B |\psi(x) f_{x_0,s}(x) - (\psi f_{x_0,s})_B|\,\mathrm{d}x \\
&\le C_\alpha \|\psi f_{x_0,s}\|_{BMO^\alpha_{\mathcal{L}}} \le C_\alpha \|f_{x_0,s}\|_{BMO^\alpha_{\mathcal{L}}} \le C.
\end{aligned}$$

The constants C and C_α appearing in this proof do not depend on $x_0 \in \mathbb{R}^n$ and $0 < s \le \frac{1}{2}\rho(x_0)$.

For the converse statement, assume ψ satisfies the properties required in the hypothesis. The kernel of the operator $T = T_\psi$ is zero and $T_\psi 1(x) = \psi(x)$, so the conclusion follows by Theorem 1.1.

The proof of (A) is completely analogous by using the function $g_{x_0,s}(x)$ of Lemma 2.5 instead of $f_{x_0,s}(x)$ and by applying Theorem 1.2. □

4 Applications

In the following subsections, we prove Theorems 1.4 and 1.3. In order to adapt our results to the applications, we need the following remark.

Remark 4.1 (Vector-valued setting) Theorems 1.1 and 1.2 can also be stated in a vector-valued setting. If Tf takes values in a Banach space $\mathbb{B}$ and the absolute values in the conditions are replaced by the norm in $\mathbb{B}$, then both results hold.

4.1 Maximal operators for the heat-diffusion semigroup $e^{-t\mathcal{L}}$.

Let $\{\mathcal{W}_t\}_{t>0}$ be the heat-diffusion semigroup associated with $\mathcal{L}$:

$$\mathcal{W}_t f(x) \equiv e^{-t\mathcal{L}} f(x) = \int_{\mathbb{R}^n} \mathcal{W}_t(x, y) f(y)\, dy, \quad f \in L^2(\mathbb{R}^n),\ x \in \mathbb{R}^n,\ t > 0.$$

The kernel of the classical heat semigroup $\{W_t\}_{t>0} = \{e^{t\Delta}\}_{t>0}$ on $\mathbb{R}^n$ is

$$W_t(x) := \frac{1}{(4\pi t)^{n/2}}\, e^{-\frac{|x|^2}{4t}}, \quad x \in \mathbb{R}^n,\ t > 0,$$

In the following arguments we need some well-known estimates about the kernel $\mathcal{W}_t(x, y)$.

Lemma 4.2 (See [14,19]) *For every $N > 0$ there exists a constant C_N such that*

$$0 \le \mathcal{W}_t(x, y) \le C_N t^{-n/2} e^{-\frac{|x-y|^2}{5t}} \left(1 + \frac{\sqrt{t}}{\rho(x)} + \frac{\sqrt{t}}{\rho(y)}\right)^{-N}, \quad x, y \in \mathbb{R}^n,\ t > 0.$$

Lemma 4.3 (See [14, Proposition 2.16]) *There exists a nonnegative function $\omega \in \mathcal{S}$ such that*

$$|\mathcal{W}_t(x, y) - W_t(x - y)| \le \left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta_0} \omega_t(x - y), \quad x, y \in \mathbb{R}^n,\ t > 0,$$

where $\omega_t(x - y) := t^{-n/2}\omega\left((x - y)/\sqrt{t}\right)$ and

$$\delta_0 := 2 - \frac{n}{q} > 0.$$

In fact, going through the proof of Dziubański and Zienkiewicz [14] we see that $\omega(x) = e^{-|x|^2}$.

Lemma 4.4 (See [13, Proposition 4.11]) *For every $0 < \delta < \delta_0$, there exists a constant $c > 0$ such that for every $N > 0$ there exists a constant $C > 0$ such that for $|y - z| < \sqrt{t}$ we have*

$$|\mathcal{W}_t(x, y) - \mathcal{W}_t(x, z)| \le C \left(\frac{|y - z|}{\sqrt{t}}\right)^{\delta} t^{-n/2}\, e^{-c|x-y|^2/t} \left(1 + \frac{\sqrt{t}}{\rho(x)} + \frac{\sqrt{t}}{\rho(y)}\right)^{-N}.$$

Lemma 4.5 (See [14, Proposition 2.17]) *For every* $0 < \delta < \min\{1, \delta_0\}$,

$$|(\mathcal{W}_t(x, y) - W_t(x - y)) - (\mathcal{W}_t(x, z) - W_t(x - z))| \le C \left(\frac{|y - z|}{\rho(x)}\right)^{\delta} \omega_t(x - y),$$

for all $x, y \in \mathbb{R}^n$ *and* $t > 0$, *with* $|y - z| < C\rho(y)$ *and* $|y - z| < \frac{1}{4}|x - y|$.

To prove that the maximal operator $\mathcal{W}^*$ defined by $\mathcal{W}^* f(x) = \sup_{t>0} |\mathcal{W}_t f(x)|$ is bounded from $BMO^{\alpha}_{\mathcal{L}}$ into itself, we give a vector-valued interpretation of the operator and apply Remark 4.1. Indeed, it is clear that $\mathcal{W}^* f = \|\mathcal{W}_t f\|_E$, with $E = L^{\infty}((0, \infty), \mathrm{d}t)$. Hence, it is enough to show that the operator $\Lambda(f) := (\mathcal{W}_t f)_{t>0}$ is bounded from $BMO^{\alpha}_{\mathcal{L}}$ into $BMO^{\alpha}_{\mathcal{L},E}$, where the space $BMO^{\alpha}_{\mathcal{L},E}$ is defined in the obvious way by replacing the absolute values $|\cdot|$ by norms $\|\cdot\|_E$.

By the Spectral Theorem, Λ is bounded from $L^2(\mathbb{R}^n)$ into $L^2_E(\mathbb{R}^n)$. The desired result is then deduced from the following proposition.

Proposition 4.6 *Let* $x, y, z \in \mathbb{R}^n$ *and* $N > 0$. *Then*

(i) $\|\mathcal{W}_t(x, y)\|_E \le \dfrac{C}{|x - y|^n} \left(1 + \dfrac{|x - y|}{\rho(x)} + \dfrac{|x - y|}{\rho(y)}\right)^{-N}$;

(ii) $\|\mathcal{W}_t(x, y) - \mathcal{W}_t(x, z)\|_E + \|\mathcal{W}_t(y, x) - \mathcal{W}_t(z, x)\|_E \le C_{\delta} \dfrac{|y - z|^{\delta}}{|x - y|^{n+\delta}}$, *whenever* $|x - y| > 2|y - z|$, *for any* $0 < \delta < 2 - \frac{n}{q}$;

(iii) *there exists a constant* C *such that for every ball* $B = B(x, s)$ *with* $0 < s \le \frac{1}{2}\rho(x)$,

$$\log\left(\frac{\rho(x)}{s}\right) \frac{1}{|B|} \int_B \|\mathcal{W}_t 1(y) - (\mathcal{W}_t 1)_B\|_E \, dy \le C,$$

and, if $\alpha < \min\{1, 2 - \frac{n}{q}\}$ *then*

$$\left(\frac{\rho(x)}{s}\right)^{\alpha} \frac{1}{|B|} \int_B \|\mathcal{W}_t 1(y) - (\mathcal{W}_t 1)_B\|_E \, dy \le C.$$

Proof Let us begin with (i). If $t > |x - y|^2$ then the conclusion is immediate from the estimate of Lemma 4.2. Assume that $t \le |x - y|^2$. Then

$$\begin{aligned}
0 \le \mathcal{W}_t(x, y) &\le \frac{C}{|x - y|^n} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(1 + \frac{\sqrt{t}}{\rho(x)} + \frac{\sqrt{t}}{\rho(y)}\right)^{-N} \\
&= \frac{C}{|x - y|^n} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(\frac{\sqrt{t}}{|x - y|}\right)^{-N} \left(\frac{|x - y|}{\sqrt{t}} + \frac{|x - y|}{\rho(x)} + \frac{|x - y|}{\rho(y)}\right)^{-N} \\
&\le \frac{C}{|x - y|^n} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(\frac{\sqrt{t}}{|x - y|}\right)^{-N} \left(1 + \frac{|x - y|}{\rho(x)} + \frac{|x - y|}{\rho(y)}\right)^{-N} \\
&\le \frac{C}{|x - y|^n} \left(1 + \frac{|x - y|}{\rho(x)} + \frac{|x - y|}{\rho(y)}\right)^{-N}.
\end{aligned}$$

We prove (ii). Observe that if $|x - y| > 2|y - z|$ then $|x - y| \sim |x - z|$. For any $0 < \delta < \delta_0$, if $|y - z| \le \sqrt{t}$, by Lemma 4.4,

$$|\mathcal{W}_t(x, y) - \mathcal{W}_t(x, z)| \le C \left(\frac{|y - z|}{\sqrt{t}}\right)^{\delta} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \le C \frac{|y - z|^{\delta}}{|x - y|^{n+\delta}}. \tag{4.1}$$

Consider the situation $|y-z|>\sqrt{t}$. Then Lemma 4.2 gives

$$|\mathcal{W}_t(x,y)| \le C\left(\frac{|y-z|}{\sqrt{t}}\right)^{\delta} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(1+\frac{\sqrt{t}}{\rho(x)}+\frac{\sqrt{t}}{\rho(y)}\right)^{-N} \le C\frac{|y-z|^{\delta}}{|x-y|^{n+\delta}}.$$

The same bound is valid for $\mathcal{W}_t(x,z)$ because $|x-z|\sim|x-y|$. Then the estimate follows directly since $|\mathcal{W}_t(x,y)-\mathcal{W}_t(x,z)| \le |\mathcal{W}_t(x,y)|+|\mathcal{W}_t(x,z)|$. The symmetry of the kernel $\mathcal{W}_t(x,y)=\mathcal{W}_t(y,x)$ gives the conclusion of (ii).

Let us prove the first statement of (iii). Let $B=B(x,s)$ with $0<s\le\frac{1}{2}\rho(x)$. The triangle inequality gives

$$\|\mathcal{W}_t 1(y)-(\mathcal{W}_t 1)_B\|_E \le \frac{1}{|B|}\int_B \|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)\|_E \,\mathrm{d}z \tag{4.2}$$

We estimate the integrand $\|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)\|_E$. Because $y,z\in B$, we have $\rho(y)\sim\rho(z)\sim\rho(x)$ (see Lemma 2.1). The fact that $W_t 1(x)\equiv 1$ and Lemma 4.3 entail

$$\begin{aligned}
|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)| &\le |\mathcal{W}_t 1(y)-W_t 1(y)|+|\mathcal{W}_t 1(z)-W_t 1(z)| \\
&\le \int_{\mathbb{R}^n}\left[\left(\frac{\sqrt{t}}{\rho(y)}\right)^{\delta_0}\omega_t(y-w)+\left(\frac{\sqrt{t}}{\rho(z)}\right)^{\delta_0}\omega_t(z-w)\right]\mathrm{d}w \\
&\le \left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta_0}\int_{\mathbb{R}^n}[\omega_t(y-w)+\omega_t(z-w)]\,\mathrm{d}w = C\left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta_0}.
\end{aligned} \tag{4.3}$$

So (4.3) gives

$$|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)| \le C\left(\frac{s}{\rho(x)}\right)^{\delta_0}, \quad \text{when } \sqrt{t}\le 2s. \tag{4.4}$$

If $\sqrt{t}>2s$ then $|y-z|\le 2s<\sqrt{t}$. Hence, Lemma 4.4 implies that

$$\begin{aligned}
|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)| &\le \int_{\mathbb{R}^n}|\mathcal{W}_t(y,w)-\mathcal{W}_t(z,w)|\,\mathrm{d}w \\
&\le C\left(\frac{|y-z|}{\sqrt{t}}\right)^{\delta} \le C\left(\frac{s}{\sqrt{t}}\right)^{\delta},
\end{aligned} \tag{4.5}$$

where $0<\delta<\delta_0$. Therefore, estimate (4.5) gives

$$|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)| \le C\left(\frac{s}{\rho(x)}\right)^{\delta}, \quad \text{when } \sqrt{t}>\rho(x). \tag{4.6}$$

When $2s<\sqrt{t}<\rho(x)$ we write

$$\begin{aligned}
|\mathcal{W}_t 1(y)-\mathcal{W}_t 1(z)| &= |(\mathcal{W}_t 1(y)-W_t 1(y))-(\mathcal{W}_t 1(z)-W_t 1(z))| \\
&= \Bigg|\left(\int_{|w-y|>C\rho(y)} + \int_{4|y-z|<|w-y|<C\rho(y)} + \int_{|w-y|<4|y-z|}\right) \\
&\quad ((\mathcal{W}_t(y,w)-W_t(y,w))-(\mathcal{W}_t(z,w)-W_t(z,w)))\,\mathrm{d}w\Bigg| \\
&= |I+II+III|.
\end{aligned}$$

For I we use the smoothness proved in part (ii) of this proposition. Note that the same smoothness estimate is valid for the classical heat kernel. So we get

$$|I| \le C \int\limits_{|w-y|>C\rho(y)} \frac{|y-z|^\delta}{|w-y|^{n+\delta}}\, \mathrm{d}w \le C \left(\frac{s}{\rho(x)}\right)^\delta.$$

In II we apply Lemma 4.5 and the fact that $\rho(w) \sim \rho(y)$ in the region of integration:

$$|II| \le C\,|y-z|^\delta \int\limits_{C\rho(y)>|w-y|>4|y-z|} \frac{\omega_t(w-y)}{\rho(w)^\delta}\, \mathrm{d}w \le C \left(\frac{s}{\rho(x)}\right)^\delta.$$

The estimate of III is obtained by applying Lemma 4.3:

$$\begin{aligned} |III| &\le C \left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta_0} \left(\int\limits_{|w-y|<4|y-z|} \omega_t(y-w)\mathrm{d}w + \int\limits_{|w-z|\le 5|y-z|} \omega_t(z-w)\mathrm{d}w \right) \\ &\le C \left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta_0} \int\limits_{|\xi|\le 5\frac{|y-z|}{\sqrt{t}}} \omega(\xi)\, \mathrm{d}\xi \le C \left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta_0} \left(\frac{|y-z|}{\sqrt{t}}\right)^n \\ &\le C \frac{s^n}{\rho(x)^{\delta_0}(\sqrt{t})^{n-\delta_0}} \le C \frac{s^n}{\rho(x)^{\delta_0} s^{n-\delta_0}} = C \left(\frac{s}{\rho(x)}\right)^{\delta_0}, \end{aligned}$$

since $2s < \sqrt{t}$ and $n - \delta_0 > 0$. Thus

$$|\mathcal{W}_t 1(y) - \mathcal{W}_t 1(z)| \le C \left(\frac{s}{\rho(x)}\right)^\delta, \quad \text{when } 2s < \sqrt{t} < \rho(x). \tag{4.7}$$

Combining (4.4), (4.6) and (4.7), we get

$$\|\mathcal{W}_t 1(y) - \mathcal{W}_t 1(z)\|_E \le C \left(\frac{s}{\rho(x)}\right)^\delta. \tag{4.8}$$

Therefore, from (4.2) and (4.8) we get

$$\log\left(\frac{\rho(x)}{s}\right) \frac{1}{|B|} \int\limits_B \|\mathcal{W}_t 1(y) - (\mathcal{W}_t 1)_B\|_E\, \mathrm{d}y \le C \left(\frac{s}{\rho(x)}\right)^\delta \log\left(\frac{\rho(x)}{s}\right) \le C,$$

which is the first conclusion of (iii).

For the second estimate of (iii), by (4.8), we have

$$\left(\frac{\rho(x)}{s}\right)^\alpha \frac{1}{|B|} \int\limits_B \|\mathcal{W}_t 1(y) - (\mathcal{W}_t 1)_B\|_E\, \mathrm{d}y \le C \left(\frac{s}{\rho(x)}\right)^{\delta-\alpha} \le C,$$

as soon as $\delta - \alpha \ge 0$, which can be guaranteed if $\alpha < \min\{1, 2 - \frac{n}{q}\}$ and we choose $\delta \ge \alpha$. □

4.2 Maximal operators for the generalized Poisson operators $\mathcal{P}_t^\sigma$.

For $0 < \sigma < 1$ we define the generalized Poisson operators $\mathcal{P}_t^\sigma$ as

$$u(x,t) \equiv \mathcal{P}_t^\sigma f(x) = \frac{t^{2\sigma}}{4^\sigma \Gamma(\sigma)} \int_0^\infty \mathrm{e}^{-\frac{t^2}{4r}} \mathcal{W}_r f(x) \, \frac{\mathrm{d}r}{r^{1+\sigma}}$$
$$= \frac{1}{\Gamma(\sigma)} \int_0^\infty \mathrm{e}^{-r} \mathcal{W}_{\frac{t^2}{4r}} f(x) \, \frac{\mathrm{d}r}{r^{1-\sigma}}, \tag{4.9}$$

for $x \in \mathbb{R}^n$ and $t > 0$. The function u satisfies the following boundary value (extension) problem:

$$\begin{cases} -\mathcal{L}_x u + \frac{1-2\sigma}{t} u_t + u_{tt} = 0, & \text{in } \mathbb{R}^n \times (0,\infty); \\ u(x,0) = f(x), & \text{on } \mathbb{R}^n. \end{cases}$$

Moreover, u is useful to characterize the fractional powers of $\mathcal{L}$ since

$$-t^{1-2\sigma} u_t(x,t)\big|_{t=0} = c_\sigma \mathcal{L}^\sigma f(x),$$

for some constant $c_\sigma > 0$, see [24]. The fractional powers $\mathcal{L}^\sigma$ can be defined in a spectral way. When $\sigma = 1/2$ we get that $\mathcal{P}_t^{1/2} = \mathrm{e}^{-t\mathcal{L}^{1/2}}$ is the classical Poisson semigroup generated by $\mathcal{L}$ given by Bochner's subordination formula, see [23]. It follows that

$$\mathcal{P}_t^\sigma f(x) = \int_{\mathbb{R}^n} \mathcal{P}_t^\sigma(x,y) f(y) \, \mathrm{d}y,$$

where

$$\mathcal{P}_t^\sigma(x,y) = \frac{t^{2\sigma}}{4^\sigma \Gamma(\sigma)} \int_0^\infty \mathrm{e}^{-\frac{t^2}{4r}} \mathcal{W}_r(x,y) \, \frac{\mathrm{d}r}{r^{1+\sigma}} \tag{4.10}$$
$$= \frac{1}{\Gamma(\sigma)} \int_0^\infty \mathrm{e}^{-r} \mathcal{W}_{\frac{t^2}{4r}}(x,y) \, \frac{\mathrm{d}r}{r^{1-\sigma}}.$$

To get the boundedness of the maximal operator

$$\mathcal{P}^{\sigma,*} f(x) := \sup_{t>0} \left|\mathcal{P}_t^\sigma f(x)\right| = \|\mathcal{P}_t^\sigma f(x)\|_E$$

in $BMO_{\mathcal{L}}^\alpha$, we proceed using the vector-valued approach and the boundedness of the maximal heat semigroup $\mathcal{W}^* f$. The following proposition completely analogous to Proposition 4.6 holds.

Proposition 4.7 *The estimates of Proposition 4.6 are valid when $\mathcal{W}_t$ is replaced by $\mathcal{P}_t^\sigma$.*

Proof The proof follows by transferring the estimates for $\mathcal{W}_t(x,y)$ to $\mathcal{P}_t^\sigma(x,y)$ through formula (4.10). We just sketch the proof of (iii). For any $y, z \in B = B(x,s)$, $x \in \mathbb{R}^n$, $0 < s \le \frac{1}{2}\rho(x)$, by (4.10), Minkowski's integral inequality and (4.8) we have

$$\|\mathcal{P}_t^\sigma 1(y) - \mathcal{P}_t^\sigma 1(z)\|_E \le C_\sigma \int_0^\infty t^{2\sigma} \mathrm{e}^{-\frac{t^2}{4r}} \|\mathcal{W}_r 1(y) - \mathcal{W}_r 1(z)\|_E \frac{\mathrm{d}r}{r^{1+\sigma}}$$
$$\le C \left(\frac{s}{\rho(x)}\right)^\delta \int_0^\infty t^{2\sigma} \mathrm{e}^{-\frac{t^2}{4r}} \frac{\mathrm{d}r}{r^{1+\sigma}} = C \left(\frac{s}{\rho(x)}\right)^\delta.$$

Then the same computations for the heat semigroup apply in this case and give (iii). □

4.3 Littlewood–Paley g-function for the heat-diffusion semigroup

The Littlewood–Paley g-function associated with $\{\mathcal{W}_t\}_{t>0}$ is defined by

$$g_{\mathcal{W}}(f)(x) = \left(\int_0^\infty |t\partial_t \mathcal{W}_t f(x)|^2 \frac{\mathrm{d}t}{t}\right)^{1/2} = \|t\partial_t \mathcal{W}_t f(x)\|_F,$$

where $F := L^2\big((0,\infty), \frac{\mathrm{d}t}{t}\big)$. The Spectral Theorem implies that $g_{\mathcal{W}}$ is an isometry on $L^2(\mathbb{R}^n)$, see Dziubański et al. [11, Lemma 3]. As before, to get the boundedness of $g_{\mathcal{W}}$ from $BMO_{\mathcal{L}}^\alpha$ into itself, it is sufficient to prove the following result.

Proposition 4.8 *The estimates of Proposition 4.6 are valid when $\mathcal{W}_t$ is replaced by $t\partial_t \mathcal{W}_t$ and the Banach space E is replaced by F.*

The proof of Proposition 4.8 requires some extra effort. Let us recall the following already well-known estimates.

Lemma 4.9 (See [11, Proposition 4]) *For any $N > 0$ there exist constants $C = C_N$ and $c > 0$ such that for all $x, y \in \mathbb{R}^n$, $t > 0$ and $0 < \delta < \delta_0$,*

(a) $|t\partial_t \mathcal{W}_t(x,y)| \le Ct^{-n/2} e^{-c\frac{|x-y|^2}{t}} \left(1 + \frac{\sqrt{t}}{\rho(x)} + \frac{\sqrt{t}}{\rho(y)}\right)^{-N}$;

(b) *For all $|h| \le \sqrt{t}$ we have*

$$|t\partial_t \mathcal{W}_t(x+h,y) - t\partial_t \mathcal{W}(x,y)| \le C \left(\frac{|h|}{\sqrt{t}}\right)^\delta \frac{e^{-c\frac{|x-y|^2}{t}}}{t^{n/2}} \left(1 + \frac{\sqrt{t}}{\rho(x)} + \frac{\sqrt{t}}{\rho(y)}\right)^{-N},$$

(c) $\left|\int_{\mathbb{R}^n} t\partial_t \mathcal{W}_t(x,y)\, dy\right| \le C \frac{(\sqrt{t}/\rho(x))^\delta}{\big(1+\sqrt{t}/\rho(x)\big)^N}.$

Proof of Proposition 4.8 Part (i) is proved using Lemma 4.9(a) and the same argument of the proof of Proposition 4.6(i).

Similarly, (ii) follows by Lemma 4.9(b) and the symmetry $\mathcal{W}_t(x,y) = \mathcal{W}_t(y,x)$.

To prove (iii) let us fix $y, z \in B = B(x_0, s)$, $0 < s \le \frac{1}{2}\rho(x_0)$. In view of an estimate like (4.2), we must handle $\|t\partial_t \mathcal{W}_t 1(y) - t\partial_t \mathcal{W}_t 1(z)\|_F$ first. We can write

$$\|t\partial_t \mathcal{W}_t 1(y) - t\partial_t \mathcal{W}_t 1(z)\|_F^2$$
$$= \int_0^\infty \left| \int_{\mathbb{R}^n} (t\partial_t \mathcal{W}_t(x,y) - t\partial_t \mathcal{W}_t(x,z))\, \mathrm{d}x \right|^2 \frac{\mathrm{d}t}{t}$$
$$= \left(\int_0^{4s^2} + \int_{4s^2}^{\rho(x_0)^2} + \int_{\rho(x_0)^2}^\infty \right) \left| \int_{\mathbb{R}^n} (t\partial_t \mathcal{W}_t(x,y) - t\partial_t \mathcal{W}_t(x,z))\, \mathrm{d}x \right|^2 \frac{\mathrm{d}t}{t}$$
$$=: A_1 + A_2 + A_3. \tag{4.11}$$

Since $y, z \in B \subset B(x_0, \rho(x_0))$, it follows that $\rho(y) \sim \rho(x_0) \sim \rho(z)$. By Lemma 4.9(c),

$$A_1 \le C \int_0^{4s^2} \frac{(\sqrt{t}/\rho(x_0))^{2\delta}}{(1+\sqrt{t}/\rho(x_0))^{2N}} \frac{\mathrm{d}t}{t}$$
$$\le C \int_0^{4s^2} \left(\frac{\sqrt{t}}{\rho(x_0)}\right)^{2\delta} \frac{\mathrm{d}t}{t} = C\left(\frac{s}{\rho(x_0)}\right)^{2\delta}. \tag{4.12}$$

Also, by Lemma 4.9(b),

$$A_3 \le C \int_{\rho(x_0)^2}^{\infty} \left(\frac{|y-z|}{\sqrt{t}}\right)^{2\delta} \left| \int_{\mathbb{R}^n} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \,\mathrm{d}x \right|^2 \frac{\mathrm{d}t}{t}$$
$$= C \int_{\rho(x_0)^2}^{\infty} \left(\frac{|y-z|}{\sqrt{t}}\right)^{2\delta} \frac{\mathrm{d}t}{t} \le C\left(\frac{s}{\rho(x_0)}\right)^{2\delta}. \tag{4.13}$$

It remains to estimate the term A_2. Recall from Dziubański et al. [11, Eq. (2.8)] that, because the potential V is in the reverse Hölder class,

$$\int_{\mathbb{R}^n} \omega_t(x-y) V(y) \,\mathrm{d}y \le \frac{C}{t}\left(\frac{\sqrt{t}}{\rho(x)}\right)^{\delta}, \quad \text{for } t \le \rho(x)^2. \tag{4.14}$$

Clearly $\partial_t \mathcal{W}_t 1(x) = \mathcal{L}\mathcal{W}_t 1(x) = \mathcal{W}_t V(x)$, that is

$$\int_{\mathbb{R}^n} \partial_t \mathcal{W}_t(x,y) \,\mathrm{d}y = \int_{\mathbb{R}^n} \mathcal{W}_t(x,y) V(y) \,\mathrm{d}y. \tag{4.15}$$

We then have, by Lemma 4.4 (remember that $|y-z| \le 2s \le \sqrt{t}$),

$$A_2 = \int_{4s^2}^{\rho(x_0)^2} \left| \int_{\mathbb{R}^n} (t\partial_t \mathcal{W}_t(x,y) - t\partial_t \mathcal{W}_t(x,z)) \,\mathrm{d}x \right|^2 \frac{\mathrm{d}t}{t}$$
$$= \int_{4s^2}^{\rho(x_0)^2} t \left| \int_{\mathbb{R}^n} (\mathcal{W}_t(y,x) - \mathcal{W}_t(z,x)) V(x) \,\mathrm{d}x \right|^2 \mathrm{d}t$$
$$\le C|y-z|^{2\delta} \int_{4s^2}^{\rho(x_0)^2} t^{1-\delta} \left| \int_{\mathbb{R}^n} t^{-n/2} \mathrm{e}^{-c\frac{|y-x|}{t}} V(x) \,\mathrm{d}x \right|^2 \mathrm{d}t$$
$$\le Cs^{2\delta} \int_{4s^2}^{\rho(x_0)^2} t^{1-\delta} t^{-2} \left(\frac{\sqrt{t}}{\rho(y)}\right)^{2\delta} \mathrm{d}t$$
$$\le C\left(\frac{s}{\rho(x_0)}\right)^{2\delta} \int_{s^2}^{\rho(x_0)^2} \frac{\mathrm{d}t}{t} = C\left(\frac{s}{\rho(x_0)}\right)^{2\delta} \log\left(\frac{\rho(x_0)}{s}\right). \tag{4.16}$$

Combining (4.11), (4.12), (4.13) and (4.16) we get

$$\|t\partial_t \mathcal{W}_t 1(y) - t\partial_t \mathcal{W}_t 1(z)\|_F \le C \left(\frac{s}{\rho(x_0)}\right)^{\delta} \left(\log\left(\frac{\rho(x_0)}{s}\right)\right)^{1/2}. \tag{4.17}$$

Thus (iii) readily follows. □

4.4 Littlewood–Paley g-function for the Poisson semigroup

The Littlewood–Paley g-function associated with the Poisson semigroup $\{\mathcal{P}_t\}_{t>0} \equiv \{\mathcal{P}_t^{1/2}\}_{t>0}$ (see (4.9) and (4.10)) is defined analogously as $g_{\mathcal{W}}$ by replacing the heat semigroup by the Poisson semigroup:

$$g_{\mathcal{P}}(f)(x) = \left(\int_0^\infty |t\partial_t \mathcal{P}_t f(x)|^2 \frac{\mathrm{d}t}{t}\right)^{1/2} = \|t\partial_t \mathcal{P}_t f(x)\|_F.$$

By the spectral Theorem, $g_{\mathcal{P}}$ is an isometry on $L^2(\mathbb{R}^n)$, see [20, Lemma 3.7]. We also have

Proposition 4.10 *The estimates of Proposition 4.6 are valid when $\mathcal{W}_t$ is replaced by $t\partial_t \mathcal{P}_t$ and the Banach space E is replaced by F.*

Proof First we derive a convenient formula to treat the operator $t\partial_t \mathcal{P}_t$. By the second identity of (4.10) with $\sigma = 1/2$ (Bochner's subordination formula) and a change of variables,

$$\begin{aligned} t\partial_t \mathcal{P}_t(x,y) &= \frac{t}{\sqrt{\pi}} \int_0^\infty \frac{\mathrm{e}^{-r}}{r^{1/2}} \partial_t \left(\mathcal{W}_{\frac{t^2}{4r}}(x,y)\right) \mathrm{d}r \\ &= \frac{t^2}{2\sqrt{\pi}} \int_0^\infty \frac{\mathrm{e}^{-r}}{r^{1/2}} \partial_v \left(\mathcal{W}_v(x,y)\right)\Big|_{v=\frac{t^2}{4r}} \frac{\mathrm{d}r}{r} \\ &= \frac{t}{\sqrt{\pi}} \int_0^\infty \mathrm{e}^{-\frac{t^2}{4v}} v\partial_v \mathcal{W}_v(x,y) \frac{dv}{v^{3/2}}. \end{aligned} \tag{4.18}$$

Formula (4.18) should be compared with the first identity of (4.10) for $\sigma = 1/2$. It will allow us to transfer the estimates for $v\partial_v \mathcal{W}_v$ to $t\partial_t \mathcal{P}_t$.

For (i) we use (4.18), Minkowski's integral inequality and the estimate for $v\partial_v \mathcal{W}_v$:

$$\begin{aligned} \|t\partial_t \mathcal{P}_t(x,y)\|_F^2 &\le C \int_0^\infty |v\partial_v \mathcal{W}_v(x,y)|^2 \int_0^\infty t e^{-\frac{t^2}{4v}} \frac{\mathrm{d}t}{t} \frac{dv}{v^{3/2}} \\ &= C \int_0^\infty |v\partial_v \mathcal{W}_v(x,y)|^2 \frac{dv}{v} \\ &\le \frac{C}{|x-y|^{2n}} \left(1 + \frac{|x-y|}{\rho(x)} + \frac{|x-y|}{\rho(y)}\right)^{-2N}. \end{aligned}$$

The estimate for (ii) follows in the same way.

By (4.18), Fubini's Theorem and (4.17),

$$\|t\partial_t \mathcal{P}_t 1(y) - t\partial_t \mathcal{P}_t 1(z)\|_F \leq C \left(\frac{s}{\rho(x_0)}\right)^{\delta} \log\left(\frac{\rho(x_0)}{s}\right)^{1/2},$$

which is sufficient for (iii). □

4.5 Laplace transform type multipliers

Given a bounded function a on $[0, \infty)$ we let

$$m(\lambda) = \lambda \int_0^\infty a(t) \mathrm{e}^{-t\lambda} \, \mathrm{d}t.$$

The Spectral Theorem allows us to define the Laplace transform type multiplier operator $m(\mathcal{L})$ associated with a, which is a bounded operator on $L^2(\mathbb{R}^n)$. Observe that

$$m(\mathcal{L}) f(x) = \int_0^\infty a(t) \mathcal{L} \mathrm{e}^{-t\mathcal{L}} f(x) \, \mathrm{d}t = \int_0^\infty a(t) \partial_t \mathcal{W}_t f(x) \, \mathrm{d}t, \quad x \in \mathbb{R}^n.$$

Then the kernel $\mathcal{M}(x, y)$ of $m(\mathcal{L})$ can be written as

$$\mathcal{M}(x, y) = \int_0^\infty a(t) \partial_t \mathcal{W}_t(x, y) \, \mathrm{d}t.$$

Proposition 4.11 *Let $x, y, z \in \mathbb{R}^n$, $N > 0$, $0 \leq \alpha < 1$ and $B = B(x, s)$ for $0 < s \leq \rho(x)$. Then*

(a) $|\mathcal{M}(x, y)| \leq \dfrac{C}{|x-y|^n} \left(1 + \dfrac{|x-y|}{\rho(x)} + \dfrac{|x-y|}{\rho(y)}\right)^{-N}$;

(b) $|\mathcal{M}(x, y) - \mathcal{M}(x, z)| + |\mathcal{M}(y, x) - \mathcal{M}(z, x)| \leq C_\delta \dfrac{|y-z|^\delta}{|x-y|^{n+\delta}}$, *for all* $|x-y| > 2|y-z|$ *and any* $0 < \delta < \delta_0$;

(c) $\log\left(\dfrac{\rho(x)}{s}\right) \dfrac{1}{|B|} \displaystyle\int_B |m(\mathcal{L})1(y) - (m(\mathcal{L})1)_B| \, dy \leq C$;

(d) $\left(\dfrac{\rho(x)}{s}\right)^\alpha \dfrac{1}{|B|} \displaystyle\int_B |m(\mathcal{L})1(y) - (m(\mathcal{L})1)_B| \, dy \leq C$, *for any* $0 \leq \alpha < \min\{1, 2 - \frac{n}{q}\}$.

Proof The reader should recall the estimates for $\partial_t \mathcal{W}_t(x, y)$ stated in Lemma 4.9.

For (a), by Lemma 4.9(a),

$$\begin{aligned}
&\int_0^{|x-y|^2} |a(t)\partial_t \mathcal{W}_t(x,y)|\ \mathrm{d}t \\
&\le C \int_0^{|x-y|^2} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(1+\frac{\sqrt{t}}{\rho(x)}+\frac{\sqrt{t}}{\rho(y)}\right)^{-N} \frac{\mathrm{d}t}{t} \\
&= C \int_0^{|x-y|^2} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(\frac{\sqrt{t}}{|x-y|}\right)^{-N} \left(\frac{|x-y|}{\sqrt{t}}+\frac{|x-y|}{\rho(x)}+\frac{|x-y|}{\rho(y)}\right)^{-N} \frac{\mathrm{d}t}{t} \\
&\le C \int_0^{|x-y|^2} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(1+\frac{|x-y|}{\rho(x)}+\frac{|x-y|}{\rho(y)}\right)^{-N} \frac{\mathrm{d}t}{t} \\
&\le \frac{C}{|x-y|^n} \left(1+\frac{|x-y|}{\rho(x)}+\frac{|x-y|}{\rho(y)}\right)^{-N},
\end{aligned}$$

and

$$\begin{aligned}
&\int_{|x-y|^2}^{\infty} |a(t)\partial_t \mathcal{W}_t(x,y)|\ \mathrm{d}t \le C \int_{|x-y|^2}^{\infty} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(1+\frac{\sqrt{t}}{\rho(x)}+\frac{\sqrt{t}}{\rho(y)}\right)^{-N} \frac{\mathrm{d}t}{t} \\
&\le C \int_{|x-y|^2}^{\infty} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \left(1+\frac{|x-y|}{\rho(x)}+\frac{|x-y|}{\rho(y)}\right)^{-N} \frac{\mathrm{d}t}{t} \\
&\le \frac{C}{|x-y|^n} \left(1+\frac{|x-y|}{\rho(x)}+\frac{|x-y|}{\rho(y)}\right)^{-N}.
\end{aligned}$$

To check (b) we apply Lemma 4.9(b) to see that

$$\begin{aligned}
&\int_{|x-y|^2}^{\infty} |a(t)|\,|\partial_t \mathcal{W}_t(x,y)-\partial_t \mathcal{W}_t(x,z)|\ \mathrm{d}t \\
&\le C \int_{|x-y|^2}^{\infty} \left(\frac{|y-z|}{\sqrt{t}}\right)^{\delta} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \frac{\mathrm{d}t}{t} \\
&\le C \frac{|y-z|^{\delta}}{|x-y|^{n+\delta}}.
\end{aligned}$$

Moreover, by Lemma 4.9(a),

$$\begin{aligned}
\int_0^{|x-y|^2} |a(t)\partial_t \mathcal{W}_t(x,y)|\ \mathrm{d}t &\le C \int_0^{|x-y|^2} \left(\frac{|y-z|}{\sqrt{t}}\right)^{\delta} t^{-n/2} \mathrm{e}^{-c\frac{|x-y|^2}{t}} \frac{\mathrm{d}t}{t} \\
&\le C \frac{|y-z|^{\delta}}{|x-y|^{n+\delta}}.
\end{aligned}$$

The same bound is valid for $\int_0^{|x-y|^2} |a(t)|\,|\partial_t \mathcal{W}_t(x,z)|\,\frac{dt}{t}$ because $|x-z| \sim |x-y|$. The symmetry of the kernel $\mathcal{M}(x,y) = \mathcal{M}(y,x)$ gives the conclusion of (b).

Fix $y, z \in B$. For (c) and (d), let us estimate the difference

$$|m(\mathcal{L})1(y) - m(\mathcal{L})1(z)| \le \|a\|_{L^\infty} \int_0^\infty \left| \int_{\mathbb{R}^n} (\partial_t \mathcal{W}_t(y,w) - \partial_t \mathcal{W}_t(z,w))\,dw \right| dt.$$

To that end we split the integral in t into three parts. We start with the part from 0 to $4s^2$. From Lemma 4.9(c),

$$\left| \int_0^{4s^2} \int_{\mathbb{R}^n} (\partial_t \mathcal{W}_t(y,w) - \partial_t \mathcal{W}_t(z,w))\,dw\,dt \right| \le C \int_0^{4s^2} \left(\frac{\sqrt{t}}{\rho(x)}\right)^\delta \frac{dt}{t} = C \left(\frac{s}{\rho(x)}\right)^\delta.$$

Let us continue with the integral from $\rho(x)^2$ to ∞. We apply Lemma 4.9(b):

$$\left| \int_{\rho(x)^2}^\infty \int_{\mathbb{R}^n} (\partial_t \mathcal{W}_t(y,w) - \partial_t \mathcal{W}_t(z,w))\;dw\,dt \right| \le C \int_{\rho(x)^2}^\infty \left(\frac{|y-z|}{\sqrt{t}}\right)^\delta \frac{dt}{t}$$
$$\le C \left(\frac{s}{\rho(x)}\right)^\delta.$$

Finally, we consider the part from $4s^2$ to $\rho(x)^2$. Applying (4.15), Lemma 4.4 and (4.14),

$$\left| \int_{4s^2}^{\rho(x)^2} \int_{\mathbb{R}^n} (\partial_t \mathcal{W}_t(y,w) - \partial_t \mathcal{W}_t(z,w))\;dw\,dt \right|$$
$$= \int_{4s^2}^{\rho(x)^2} \left| \int_{\mathbb{R}^n} (\mathcal{W}_t(y,w) - \mathcal{W}_t(z,w))\,V(w)\,dw \right| dt$$
$$\le C\,|y-z|^\delta \int_{4s^2}^{\rho(x)^2} \int_{\mathbb{R}^n} t^{-n/2} e^{-c\frac{|y-w|^2}{t}} V(w)\,dw\,\frac{dt}{t^{\delta/2}}$$
$$\le C \left(\frac{s}{\rho(y)}\right)^\delta \int_{s^2}^{\rho(x)^2} \frac{dt}{t} \le C \left(\frac{s}{\rho(x)}\right)^\delta \log\left(\frac{\rho(x)}{s}\right).$$

Hence,

$$\frac{1}{|B|} \int_B |m(\mathcal{L})1(y) - (m(\mathcal{L})1)_B|\;dy \le \frac{C}{s^{2n}} \int_B \int_B |m(\mathcal{L})1(y) - m(\mathcal{L})(z)|\,dy\,dz$$
$$\le C \left(\frac{s}{\rho(x)}\right)^\delta \log\left(\frac{\rho(x)}{s}\right).$$

Thus, (c) is valid and also (d) holds when $\alpha < \delta$. □

4.6 Riesz transforms

For every $i = 1, 2, \ldots, n$, the i-th Riesz transform $\mathcal{R}_i$ associated with $\mathcal{L}$ is defined by

$$\mathcal{R}_i = \partial_{x_i} \mathcal{L}^{-1/2} = \partial_{x_i} \frac{1}{\sqrt{\pi}} \int_0^\infty \mathrm{e}^{-t\mathcal{L}} \frac{\mathrm{d}t}{t^{1/2}}.$$

We denote by $\mathcal{R}$ the vector $\nabla \mathcal{L}^{-1/2} = (\mathcal{R}_1, \ldots, \mathcal{R}_n)$. The Riesz transforms associated with $\mathcal{L}$ were first studied by Shen in [22]. He showed (Theorem 0.8 of [22]) that if the potential $V \in RH_q$ with $q > n$, then $\mathcal{R}$ is a Calderón–Zygmund operator. In particular, the $\mathbb{R}^n$-valued operator $\mathcal{R}$ is bounded from $L^2(\mathbb{R}^n)$ into $L^2_{\mathbb{R}^n}(\mathbb{R}^n)$ and its kernel $\mathcal{K}$ satisfies, for any $0 < \delta < 1 - \frac{n}{q}$,

$$|\mathcal{K}(x, y) - \mathcal{K}(x, z)| + |\mathcal{K}(y, x) - \mathcal{K}(z, x)| \le C \frac{|y - z|^\delta}{|x - y|^{n+\delta}}, \tag{4.19}$$

whenever $|x - y| > 2|y - z|$. Moreover, when $q > n$ we have that for any $x, y \in \mathbb{R}^n$, $x \neq y$, and $N > 0$ there exists a constant C_N such that

$$|\mathcal{K}(x, y)| \le \frac{C_N}{|x - y|^n} \left(1 + \frac{|x - y|}{\rho(x)}\right)^{-N}, \tag{4.20}$$

see [22, Eq. (6.5)] and also [5, Lemma 3]. Hence, $\mathcal{R}$ is a γ-Schrödinger–Calderón–Zygmuund operator with $\gamma = 0$.

The boundedness results of $\mathcal{R}$ in $BMO^\alpha_{\mathcal{L}}$ follow by checking the properties of $\mathcal{R}1$.

Proposition 4.12 *Let $V \in RH_q$ with $q > n$ and $B = B(x_0, s)$ for $x_0 \in \mathbb{R}^n$ and $0 < s \le \frac{1}{2}\rho(x_0)$. Then*

(i) $\log\left(\frac{\rho(x_0)}{s}\right) \frac{1}{|B|} \int_B |\mathcal{R}1(y) - (\mathcal{R}1)_B| \, dy \le C$;

(ii) $\left(\frac{\rho(x_0)}{s}\right)^\alpha \frac{1}{|B|} \int_B |\mathcal{R}1(y) - (\mathcal{R}1)_B| \, dy \le C$, *for* $\alpha < 1 - \frac{n}{q}$.

To prove Proposition 4.12, we collect some well-known estimates on $\mathcal{K}(x, y)$. Let us denote by $\mathcal{K}_0$ the kernel of the ($\mathbb{R}^n$-valued) classical Riesz transform $\mathcal{R}_0 = \nabla(-\Delta)^{-1/2}$.

Lemma 4.13 ([5, Lemmas 3 and 4]) *Suppose that $V \in RH_q$ with $q > n$.*

(a) *For any $x, y \in \mathbb{R}^n$, $x \neq y$,*

$$|\mathcal{K}(x, y) - \mathcal{K}_0(x, y)| \le \frac{C}{|x - y|^n} \left(\frac{|x - y|}{\rho(x)}\right)^{2-n/q}.$$

(b) *For any $0 < \delta < 1 - \frac{n}{q}$ there exists a constant C such that if $|z - y| \ge 2|x - y|$ then*

$$|(\mathcal{K}(x, z) - \mathcal{K}_0(x, z)) - (\mathcal{K}(y, z) - \mathcal{K}_0(y, z))| \le C \frac{|x - y|^\delta}{|z - y|^{n+\delta}} \left(\frac{|z - y|}{\rho(z)}\right)^{2-n/q}.$$

Proof of Proposition 4.12 Let $y, z \in B$. Then $\rho(y) \sim \rho(x_0) \sim \rho(z)$. Since

$$\mathcal{R}1(x) = \lim_{\varepsilon \to 0^+} \int_{|x-y|>\varepsilon} \mathcal{K}(x, y) \, \mathrm{d}y, \quad \text{a.e. } x \in \mathbb{R}^n,$$

we have

$$
\begin{aligned}
|\mathcal{R}1(y)-\mathcal{R}1(z)| \le & \lim_{\varepsilon\to 0^+}\left|\int\limits_{\varepsilon<|x-y|\le 4\rho(x_0)} \mathcal{K}(y,x)\,\mathrm{d}x - \int\limits_{\varepsilon<|x-z|\le 4\rho(x_0)} \mathcal{K}(z,x)\,\mathrm{d}x\right| \\
& + \left|\int\limits_{|x-y|>4\rho(x_0)} \mathcal{K}(y,x)\,\mathrm{d}x - \int\limits_{|x-z|>4\rho(x_0)} \mathcal{K}(z,x)\,\mathrm{d}x\right| \\
=: & \lim_{\varepsilon\to 0^+} A_\varepsilon + B.
\end{aligned}
$$

First, let us consider A_ε. Since we will consider the limit as ε tends to zero, we can assume that $0<\varepsilon<4\rho(x_0)-2s$. For every annulus E we have $\int_E \mathcal{K}_0(x,y)\,\mathrm{d}y=0$. Therefore,

$$
\begin{aligned}
A_\varepsilon = & \left|\int\limits_{\varepsilon<|x-y|\le 4\rho(x_0)} (\mathcal{K}(y,x)-\mathcal{K}_0(y,x))\,\mathrm{d}x \right.\\
& \left. - \int\limits_{\varepsilon<|x-z|\le 4\rho(x_0)} (\mathcal{K}(z,x)-\mathcal{K}_0(z,x))\,\mathrm{d}x\right| \\
\le & \left|\int\limits_{\mathbb{R}^n} (\mathcal{K}(y,x)-\mathcal{K}_0(y,x))\left(\chi_{\varepsilon<|x-y|\le 4\rho(x_0)}(x)-\chi_{\varepsilon<|x-z|\le 4\rho(x_0)}(x)\right)\mathrm{d}x\right| \\
& + \left|\int\limits_{\mathbb{R}^n} [(\mathcal{K}(y,x)-\mathcal{K}_0(y,x))-(\mathcal{K}(z,x)-\mathcal{K}_0(z,x))]\,\chi_{\varepsilon<|x-z|\le 4\rho(x_0)}(x)\mathrm{d}x\right| \\
=: & A_\varepsilon^1 + A_\varepsilon^2. \qquad (4.21)
\end{aligned}
$$

The term A_ε^1 is not zero when $\left|\chi_{\varepsilon<|x-y|\le 4\rho(x_0)}(x)-\chi_{\varepsilon<|x-z|\le 4\rho(x_0)}(x)\right|=1$, namely, when

- $\varepsilon<|x-y|\le 4\rho(x_0)$ and $|x-z|\le\varepsilon$; or
- $\varepsilon<|x-y|\le 4\rho(x_0)$ and $|x-z|>4\rho(x_0)$; or
- $\varepsilon<|x-z|\le 4\rho(x_0)$ and $|x-y|\le\varepsilon$; or
- $\varepsilon<|x-z|\le 4\rho(x_0)$ and $|x-y|>4\rho(x_0)$.

In the first case we have $\varepsilon<|x-y|\le|x-z|+|z-y|<\varepsilon+2s$. Then, by Lemma 4.13(a),

$$
A_\varepsilon^1 \le \int\limits_{\varepsilon<|x-y|\le 2s+\varepsilon} \frac{C}{|x-y|^n}\left(\frac{|x-y|}{\rho(y)}\right)^{2-n/q}\mathrm{d}x \le C\left(\frac{s}{\rho(x_0)}\right)^{2-n/q}. \qquad (4.22)
$$

In the second case, by the assumption on ε, we get $\max\{\varepsilon, 4\rho(x_0)-2s\}=4\rho(x_0)-2s<|x-y|\le 4\rho(x_0)$. Then Lemma 4.13(a) and the Mean Value Theorem give

$$
A_\varepsilon^1 \le \frac{C}{\rho(x_0)^{2-n/q}} \int\limits_{4\rho(x_0)-2s<|x-y|\le 4\rho(x_0)} |x-y|^{2-n/q-n}\,\mathrm{d}x \le C\frac{s}{\rho(x_0)}. \qquad (4.23)
$$

In the third and fourth cases we obtain the same bounds as in (4.22) and (4.23) by replacing y by z. Thus, when $0 < \delta < 1 - n/q$,

$$A_\varepsilon^1 \le C\left(\frac{s}{\rho(x_0)}\right)^\delta. \tag{4.24}$$

We see that A_ε^2 is bounded by $\left|A_\varepsilon^{2,1}\right| + \left|A_\varepsilon^{2,2}\right|$, where

$$\begin{aligned} A_\varepsilon^{2,1} + A_\varepsilon^{2,2} = &\int\limits_{|x-z|>2|y-z|} [(\mathcal{K}(y,x) - \mathcal{K}_0(y,x)) - (\mathcal{K}(z,x) - \mathcal{K}_0(z,x))] \\ &\qquad \times \chi_{\varepsilon<|x-z|\le 4\rho(x_0)}(x)\mathrm{d}x \\ &+ \int\limits_{|x-z|\le 2|y-z|} [(\mathcal{K}(y,x) - \mathcal{K}_0(y,x)) - (\mathcal{K}(z,x) - \mathcal{K}_0(z,x))] \\ &\qquad \times \chi_{\varepsilon<|x-z|\le 4\rho(x_0)}(x)\mathrm{d}x. \end{aligned} \tag{4.25}$$

By Lemma 4.13(b),

$$A_\varepsilon^{2,1} \le C\frac{|y-z|^\delta}{\rho(z)^{2-n/q}} \int\limits_{|x-z|\le 4\rho(x_0)} |x-z|^{2-n/q-n-\delta}\,\mathrm{d}x \le C\left(\frac{s}{\rho(x_0)}\right)^\delta. \tag{4.26}$$

On the other hand, Lemma 4.13(a) gives

$$\begin{aligned} A_\varepsilon^{2,2} &\le \int\limits_{|x-z|\le 2|y-z|} \frac{C}{|x-y|^n}\left(\frac{|x-y|}{\rho(y)}\right)^{2-n/q}\mathrm{d}x \\ &\quad + \int\limits_{|x-z|\le 2|y-z|} \frac{C}{|x-z|^n}\left(\frac{|x-z|}{\rho(z)}\right)^{2-n/q}\mathrm{d}x \\ &\le \frac{C}{\rho(x_0)^{2-n/q}} \int\limits_{|x-y|\le 3|y-z|} |x-y|^{2-n/q-n}\,\mathrm{d}x \\ &\quad + \frac{C}{\rho(x_0)^{2-n/q}} \int\limits_{|x-z|\le 2|y-z|} |x-z|^{2-n/q-n}\,\mathrm{d}x \\ &\le C\left(\frac{s}{\rho(x_0)}\right)^{2-n/q} \le C\left(\frac{s}{\rho(x_0)}\right)^\delta, \end{aligned} \tag{4.27}$$

for any $0 < \delta < 1 - n/q$. Hence, from (4.21), (4.24), (4.25), (4.26) and (4.27) we obtain that for all $\varepsilon > 0$ sufficiently small,

$$A_\varepsilon \le C\left(\frac{s}{\rho(x_0)}\right)^\delta. \tag{4.28}$$

Let us now estimate B. In a similar way,

$$\begin{aligned} B &\leq \int\limits_{|x-y|>4\rho(x_0)} |\mathcal{K}(y,x)-\mathcal{K}(z,x)|\ \mathrm{d}x \\ &\quad + \int\limits_{\mathbb{R}^n} |\mathcal{K}(z,x)|\left|\chi_{|x-z|>4\rho(x_0)}(x)-\chi_{|x-z|>4\rho(x_0)}(x)\right|\ \mathrm{d}x \\ &=: B_1+B_2. \end{aligned}$$

In the integrand of B_1 we have $|x-y|>4\rho(x_0)\geq 8s>2\,|y-z|$. Therefore, the smoothness of the Riesz kernel (4.19) can be applied to get

$$B_1 \leq C \int\limits_{|x-y|>4\rho(x_0)} \frac{|y-z|^\delta}{|x-y|^{n+\delta}}\ \mathrm{d}x \leq C\left(\frac{s}{\rho(x_0)}\right)^\delta.$$

It is possible to deal with B_2 as with A_ε^1 above to derive the same bound. Hence,

$$B \leq C\left(\frac{s}{\rho(x_0)}\right)^\delta.$$

This last estimate together with (4.28) implies

$$|\mathcal{R}1(y)-\mathcal{R}1(z)| \leq C\left(\frac{s}{\rho(x_0)}\right)^\delta,$$

where $0<\delta<1-n/q$. From here (i) and (ii) readily follow. □

4.7 Negative powers

For any $\gamma>0$ the negative powers of $\mathcal{L}$ are defined as

$$\mathcal{L}^{-\gamma/2} f(x) = \frac{1}{\Gamma(\gamma/2)} \int\limits_0^\infty \mathrm{e}^{-t\mathcal{L}} f(x)\ \frac{\mathrm{d}t}{t^{1-\gamma/2}} = \int\limits_{\mathbb{R}^n} \mathcal{K}_\gamma(x,y) f(y)\ \mathrm{d}y,$$

where

$$\mathcal{K}_\gamma(x,y) = \frac{1}{\Gamma(\gamma/2)} \int\limits_0^\infty \mathcal{W}_t(x,y)\ \frac{\mathrm{d}t}{t^{1-\gamma/2}}, \quad x\in\mathbb{R}^n.$$

Therefore, by Lemma 4.2 and a similar argument as in the proof of Proposition 4.6(i), for every $N>0$,

$$\left|\mathcal{K}_\gamma(x,y)\right| \leq \frac{C}{|x-y|^{n-\gamma}}\left(1+\frac{|x-y|}{\rho(x)}+\frac{|x-y|}{\rho(y)}\right)^{-N}.$$

In particular, $\mathcal{L}^{-\gamma/2}$ is bounded from $L^p(\mathbb{R}^n)$ into $L^q(\mathbb{R}^n)$, for $\frac{1}{q}=\frac{1}{p}-\frac{\gamma}{n}$ with $1<p<q<\infty$ and $0<\gamma<n$. Using similar arguments to those in the proof of Proposition 4.6(ii), it can be checked that

$$\left|\mathcal{K}_\gamma(x,y)-\mathcal{K}_\gamma(x,z)\right| + \left|\mathcal{K}_\gamma(y,x)-\mathcal{K}_\gamma(z,x)\right| \leq C\frac{|y-z|^\delta}{|x-y|^{n-\gamma+\delta}},$$

when $|x-y| > 2|y-z|$, for any $0 < \delta < 2 - \frac{n}{q}$. Thus, $\mathcal{L}^{-\gamma}$ is a γ-Schrödinger–Calderón–Zygmund operator according to Definition 3.1.

The second item of Theorem 1.4 is a consequence of the following proposition and our two main theorems.

Proposition 4.14 *Let $B = B(x,s)$ with $0 < s \le \frac{1}{2}\rho(x)$. Then*

(i) $\log\left(\frac{\rho(x)}{s}\right)\frac{1}{|B|^{1+\frac{\gamma}{n}}}\int_B |\mathcal{L}^{-\gamma/2}1(y) - (\mathcal{L}^{-\gamma/2}1)_B|\,dy \le C$ *if* $\gamma \le 2 - \frac{n}{q}$;

(ii) $\left(\frac{\rho(x)}{s}\right)^{\alpha}\frac{1}{|B|^{1+\frac{\gamma}{n}}}\int_B |\mathcal{L}^{-\gamma/2}1(y) - (\mathcal{L}^{-\gamma/2}1)_B|\,dy \le C$ *if* $\alpha + \gamma < \min\{1, 2 - \frac{n}{q}\}$.

Proof Fix $y, z \in B$, so that $\rho(x) \sim \rho(y) \sim \rho(z)$. We can write

$$\mathcal{L}^{-\gamma/2}1(y) - \mathcal{L}^{-\gamma/2}1(z) = \int_0^\infty \int_{\mathbb{R}^n} (\mathcal{W}_t(y,w) - \mathcal{W}_t(z,w))\,\mathrm{d}w\, t^{\gamma/2}\,\frac{\mathrm{d}t}{t}. \tag{4.29}$$

We split the integral in t of the difference (4.29) into two parts. From (4.8) we have

$$\left|\int_0^{\rho(x)^2}\int_{\mathbb{R}^n} (\mathcal{W}_t(y,w) - \mathcal{W}_t(z,w))\,\mathrm{d}w\, t^{\gamma/2}\,\frac{\mathrm{d}t}{t}\right|$$
$$\le C\left(\frac{s}{\rho(x)}\right)^{\delta}\int_0^{\rho(x)^2} t^{\gamma/2}\,\frac{\mathrm{d}t}{t} = C\left(\frac{s}{\rho(x)}\right)^{\delta}\rho(x)^{\gamma}.$$

On the other hand, we can use (4.5) to get

$$\left|\int_{\rho(x)^2}^\infty \int_{\mathbb{R}^n} (\mathcal{W}_t(y,w) - \mathcal{W}_t(z,w))\,\mathrm{d}w\, t^{\gamma/2}\,\frac{\mathrm{d}t}{t}\right|$$
$$\le C\int_{\rho(x)^2}^\infty \left(\frac{s}{\sqrt{t}}\right)^{\delta} t^{\gamma/2}\,\frac{\mathrm{d}t}{t} \le C\left(\frac{s}{\rho(x)}\right)^{\delta}\rho(x)^{\gamma},$$

since $\gamma < \delta$. An application of these last two estimates to (4.29) finally gives

$$\frac{1}{|B|^{1+\frac{\gamma}{n}}}\int_B |\mathcal{L}^{-\gamma/2}1(y) - (\mathcal{L}^{-\gamma/2}1)_B|\mathrm{d}y$$
$$\le \frac{C}{s^{2n+\gamma}}\int_B\int_B |\mathcal{L}^{-\gamma/2}1(y) - \mathcal{L}^{-\gamma/2}1(z)|\,\mathrm{d}y\,\mathrm{d}z \le C\left(\frac{s}{\rho(x)}\right)^{\delta-\gamma}.$$

Thus, (i) is valid if $\gamma < 2 - \frac{n}{q}$ and $\delta < 2 - \frac{n}{q}$ is chosen such that $\gamma \le \delta$. Also (ii) holds when $\alpha + \gamma < \min\{1, 2 - \frac{n}{q}\}$. □

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